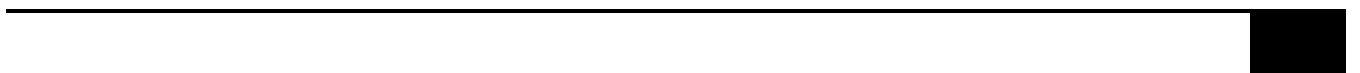


Canon EF Protocol Reverse Engineering Summary

Draft 1 – Chris E – 8/26/2026



Revision History

Date	Revision	Author	Notes
8/26/2026	Draft 1	Chris E	Initial Draft Release

DRAFT

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Forward and Acknowledgements

This document serves as a summary of findings obtained through personal reverse engineering efforts. The legality of reverse engineering methods varies from country to country, readers should check their local regulations. Several findings here were originally discovered and documented on the website <http://jp79dsfr.free.fr/Photo-Tech/index.htm> (JP, Photo-Tech, 2016). Any information replicated in this document was independently verified and replicated through my own work where possible, and marked as not verified where not possible. For the rest of the document I will refer to the author as JP and mark reused data.

The aim of this document is to show how the information was arrived at, and how it was used to develop a personal project aimed at producing something that would be interoperable with Canon EF cameras. No information was arrived at using original manufacture data. Information is presented only for general purpose use. It should also be understood that the information here may be incomplete and there is no guarantee that it will allow for full interoperability with any and all Canon lenses or camera bodies.

Note: The information in this document is incomplete as of this Draft revision, the document will be updated as more information is worked out

Technical Definitions

Protocol analysis and data encoding are all new subjects to me and I have had to learn it all on the fly while working on this project. As it is possible that I have used terminology incorrectly, this section serves to outline technical terms as I have used them. I will work to correct any mistakes as they are discovered.

Binary Encoding and Hexadecimal

Since the Canon EF protocol is a modified SPI bus, the data it sends is formatted as hexadecimal. The data being sent follows assorted methods for encoding, depending on what works best for specific parameters.

Terms used in this document

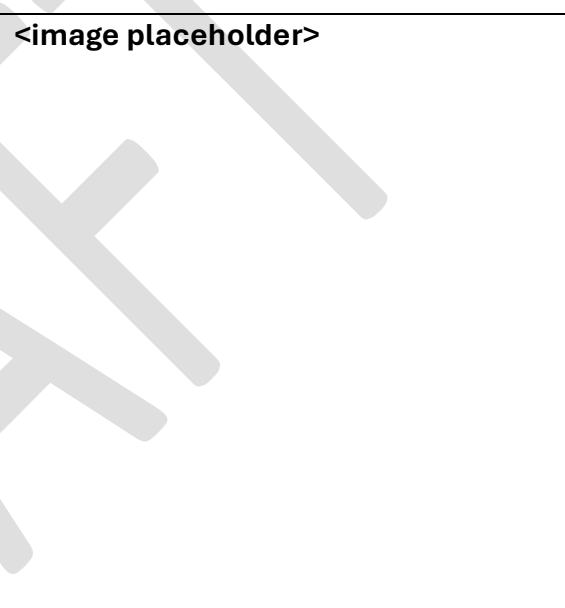
Term	Meaning	Example
Bit	Single binary digit	0 / 1
Nibble	4 bits, one Hexadecimal Digit	1010 = 0xA
Byte	8 bits, two hexadecimal digits, decimal range of 0-255	0x7F
Word	16 bits, four hexadecimal digits, decimal range of 0-65535	0x7FFF
Dword/Long	Any word longer than 16 bits, in 4 bit multiples	0x0000FFFF

EF Physical Mount

A camera mounting system can be broken into two parts. First is the physical retention mechanism that holds a lens onto a camera body. Second is the electrical interface that allows for communication between the camera and lens.

The mechanical Drawing for the EF mount is attached in Appendix C – Mechanical Drawings

The EF mount uses 8 pins, laid out as shown.

#	Function	Description	<image placeholder>
1	VBAT	Power Supply for lens motors	
2	Detect	Lens Presence Detection	
3	P-Ground	Ground Pin for lens motors	
4	VDD	Logic power for Lens controls and data	
5	Data Camera	Data line from Camera to Lens	
6	Data Lens	Data Lin from Lens to Camera	
7	Clock	Clock Line for Data Communication	
8	D-Ground	Ground Pin for data interface	

Additionally, some lenses may contain an extra 3 pins, used when a teleconverter is present. No testing has been completed by me with a teleconverter at this time. If I am able to buy one at a later point I will update with more information on the teleconverter pins.

Power

VBAT – 4.75V

This is the power used to drive all motors in lenses. It is generated by a dc-dc converter in the camera. In early film cameras this pin was directly connected to battery and provided a higher and variable voltage. Power is only available during times when it is expected that motors will be need to run, otherwise it is shut off.

VDD – 5V

This is the power supplied to run the logic and control circuits in the lens. It is enabled when a lens is connected by the lens shorting the Detect and P-Ground pins together. This power is enabled at all times as long as a lens is connected.

Grounds

P-ground and D-ground exist at the same level, but are connected to different places in the camera for EMI reasons. It is important to maintain some separation between these when designing a controller that connects to the camera mount.

Detect

This is a signal line to the camera. It has a pull up resistor to 5V. When a lens connects it shorts this pin to P-ground, which serves as a signal to turn on the VDD power.

Data Signals (Clock, Data Camera, Data Lens)

These are signal lines used for data communication between the camera and lens. They are 5V logic and idle high.

How Mirrorless Lenses Detect EF Lenses

I plan on eventually repeating this work with the EF-m and RF mounts, so I do not want to go into too much detail on those mounts in this document. However I think it's important to explain how mirrorless Canon Cameras detect the older EF lenses since they are the most likely candidates to use this work.

EF-M Cameras

Much of the information here was learned from [pointsinfocus.com](https://www.pointsinfocus.com) (Franke, 2018), where there is an article going over how the EF-M mount handles detection. It's a very well written article and you should go read it:

<https://www.pointsinfocus.com/learning/cameras-lenses/canon-ef-m-mount-lens-type-detection/>

The short version is that the EF-m mount added two new pins called MIF and DETEF. The MIF pin is used to detect the presence of a lens, and the DETEF pin is used to determine what kind of lens is attached.

While EF lenses operate at 5V logic, EF-m lenses operate at 3.3V logic, so the camera needs some way to determine what logic level to use, and what protocol to expect. This needs to be fairly fool proof so that 5v logic isn't sent to the EF-m lenses as that would damage the microcontroller.

When no lens is attached all pins are deenergized except for MIF and DETEF. MIF is pulled high internally, and when a lens is fully attached it connects that pin to ground which signals that a lens is present.

DETEF makes use of a design characteristic of the EF lens mount. On EF and EF-s lenses, the main power ground for the motors is a double wide pad that interfaces with two pins in the lens mount. In the EF to EF-m adapter, these two pins are routed to two different places. One connects directly to power ground, and the other connects to DETEF. Internal to the camera the DETEF pin is pulled high through a 100k resistor.

When an EF lens is attached, it pulls the DETEF pin completely to ground. EF-m lenses however have an internal 300k resistor connecting the DETEF pin to ground, which results in a voltage divider circuit feeding some voltage to the camera's microcontroller.

The result is the ability to know what kind of lens has been attached, which allows the camera to select the appropriate logic level to enable.

The EF to EF-m adapter pin mapping can be seen in Appendix A – Testing Hardware Schematics

RF cameras

The RF mount appears to use a very similar method to the EF-m cameras. There is again a pin that appears to be used as a detect type pin. The EF to RF adapter again splits the two power ground pins and this time routes one of them through a 1k resistor to this pin. It looks like the decision was made to require EF lenses to participate in the voltage divider in some way.

The EF to RF adapter pin mapping can be seen in Appendix A – Testing Hardware Schematics

Autofocus Theory of Operation

Understanding the basic theory of autofocus makes understanding the EF protocol a little easier, it explains the flow of commands and the information being communicated. At the very basic level the autofocus loop is simple, measure the amount of defocus, issue a command to move the lens' focus unit to correct that defocus. The challenging part is in making sure you have all the information to make that calculation.

Defocus Measurement

Defocus is a physical measurement of how far in front of or behind the sensor plane the focusing plane of the lens currently is. In DSLR cameras this was done with a dedicated sensor. Light heading to the viewfinder was also split and directed to this sensor, which could calculate how far apart the split light was landing and use that to calculate defocus.

Mirrorless cameras however do not reroute the light, and instead use specialized pixels on the sensor that can perform the same function.

There are far better articles that explain the functionality of these sensors, I would encourage those curious to know more to seek them out.

Defocus Correction

Calculating the defocus correction is easy at first glance. It's basically:

$$\text{Required Optical Movement Distance} = \text{Measured Defocus} * \text{Sensitivity Coefficient}$$

What's complicated is that sensitivity coefficient, because it changes based on the lens, that lens' design, and often the settings you're using on the camera. Calculating that coefficient works mostly in the following loop:

1. Start with a base sensitivity. This will be specific to each lens, but it's a ratio for how far the focus plane moves relative to the sensor compared to how far the focusing unit in the lens has physically moved.
2. Adjust sensitivity with any local adjustments that the lens provides, typically some kind of ratio based on desired focus direction so that move size scales with the rate that sensitivity changes in that direction.
3. Adjust sensitivity further based on any lens specific compensation. This might be minor adjustments based focus distance, zoom distance, or factors like the aperture setting and optical design.
4. Map this adjusted coefficient to the focus drive of the lens using the provided distance move per step value provided by the lens.

5. Send that command to the lens.
6. Measure defocus again and repeat.

The earliest versions of the EF protocol use this base set of corrections. As the protocol was updated, they added more parameters for finer correction and performance, but the basic loop remains the same.

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EF protocol analysis

General information

The Canon EF protocol is based on the Serial Peripheral Interface (SPI). While the protocol is based on SPI, it has been customized to meet Canon's needs making most common SPI libraries and drivers fail. The deviations from standard SPI are outlined in their own section.

This document will not go over the basics of SPI, there are plenty of good resources that cover the subject. If you'd like to get up to speed I can recommend the following two articles, though there are plenty more.

<https://learn.sparkfun.com/tutorials/serial-peripheral-interface-spi/all>

<https://www.analog.com/en/resources/analog-dialogue/articles/introduction-to-spi-interface.html>

Important Note

The way SPI works is that the camera and lens send data at the same time, however the data being sent by the lens is the answer to the command sent in the previous frame from the camera. So, when reviewing the data captures and command tables it is important to remember that the value in the Lens column corresponds to the camera command from the preceding line/frame.

A word on Nomenclature

Historically the data lines used in SPI were referred to as MOSI and MISO which stand for master out/slave in and master in/slave out. These terms can be viewed as problematic and in more recent years there has been a push to change the terminology. Some documentation has shifted to using main and subnode/ peripheral, others have replaced the acronym completely with PICO/POCI or COPI/CIPO using controller/peripheral. For the sake of this document I will be calling out the lines with their explicit use as “camera to lens” and “lens to camera”.

Technical Details and Structure

The protocol is based on the Motorola SPI Mode 3 standard (Signals Idle high, Data is set on the falling edge and read on the rising edge), 8bits MSB first.

The EF protocol has undergone several updates over the years resulting in several “generations” each with additional capabilities added. In order to retain backwards compatibility the camera and lens always begin communication in the oldest version of the

protocol and negotiate the use of a newer version. However, the technical details of the protocol remain largely the same for all generations. The largest change is the operating speed. Any mention of timing in this section will list the timing for each generation. The original version of the protocol operates at roughly 80kHz, while the newer versions operate at roughly 500kHz.

Deviations from Standard SPI

The largest deviation from standard SPI is the fact that the lens is expected to acknowledge each byte by pulling on the clock line. In standard SPI only the controller is allowed to toggle the clock line. In addition, the length of this acknowledge can be variable, allowing the lens to indicate “busy”. This is very similar to clock stretching with I2C.

Seen in the following figure, there is a fixed length of time at clock idle where the camera and lens swap roles, then the low pulse from the lens, and then some variable idle time before the next byte starts.

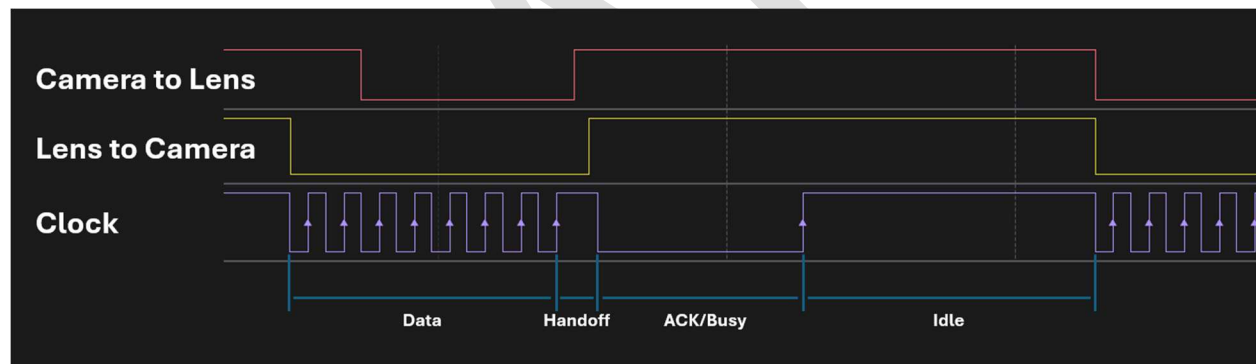


Figure 1: Data Frame Acknowledgement Behavior

Timing at 80 kHz

- Data Frame: ~90 μ S (~6 μ S/pulse)
- Handoff: ~15 μ S
- ACK/Busy: 65 μ S min, stretchable
- Idle: 100 μ S min, stretchable

Timing at 500 kHz

- Data Frame: ~16 μ S (~1 μ S/pulse)
- Handoff: ~10 μ S
- ACK/Busy: 11 μ S min, stretchable
- Idle: 11 μ S min, stretchable

Protocol Generations

There appears to be 3 generations of the EF protocol, with each generation adding new features. While I do not know for sure when each generation was officially put into use, I can make some educated guesses.

Generation 1 – Film SLRs

Generation 1 would be the first iteration of the EF protocol, written and implemented in the last 1980's for the first EF SLR film cameras like the EOS 1. This generation is the most simplistic, containing only the core information required to allow the camera to calculate focus moves and basic lens information.

Generation 2 – DSLRs

Generation 2 seems to have been implemented with the first DSLR models. This generation brought more granular information to the lenses in the form of focus distance encoders and more localized sensitivity correction values to better account for the changes that occur with zoom and focus distance. This is also where the faster 500kHz communication rate was introduced, allowing for faster focus and better tracking.

Generation 3 – Live View and Mirrorless

Generation 3 looks to be implemented around the time that DSLR cameras started having live view with sensor based focus control rather than the phase shift only focus control. This would have lead into the mirrorless camera generation which moved exclusively to sensor based auto focus. This generation brought even more correction commands the lens could send, and also shuffled some commands to new values.

Interestingly, the later EF-m cameras seem to largely use this generation 3 protocol with only a few small changes to timing. So fully decoding the generation 3 protocol should unlock EF-m cameras and lenses as well.

Typical Communication Order of Operations and sequencing.

Lens Wake/Initialization

The camera always follows a set order of commands to initialize communication with a lens. This happens both when the camera is switched off and a lens is attached, when the camera is switched on, and when the camera comes out of standby. This serves as a check to make sure that the lens hasn't been changed since the last handshake.

The camera sends some number of null 0x00 commands to see if the lens is responding. The lens will raise the Lens to Camera line to idle high once it is powered on and then depending on the specific lens it will either reply with 0x00, or 0xFF to these initial null commands.

The camera then starts sending the 0x0A availability check, where a reply from the lens of 0xAA indicates it is present and ready. The camera sends this command several times in a row until it is satisfied that the lens is truly present.

The camera then sends the 0x80 command to learn what lens was attached. This tells the camera the lens type, lens ID, focal range, and then what features the lens has. If the lens says that is capable of the new generation(s) of protocol the camera and lens swap to the faster speed at this time. I will split the rest of the sequence from here based on protocol.

Generation 1 Protocol

Following the 0x80 command the camera sends several initialization commands. It sends an initial AF speed setting with the 0x50 command, resets the autofocus counter with 0xC0, initializes aperture with an initial move to home command through 0x13, and it begins repeatedly polling AV/AF status with 0x90.

At this point, if the camera is in the off position or in standby it sends the sleep command. If it is switched to on and active it enters the main loop outlined below.

Generation 2 and 3 Protocol

Technically the camera and lens switch speed in the middle of the 0x80 command sequence. It appears that as soon as the lens says it is capable, the camera switches. Following the completion of the 0x80 command sequence the camera repeats the initialization from the original protocol, this time with some new commands. On the 3rd generation of the protocol it issues a series of 0x8n and some 0xCn commands. It then does an initial poll of all the autofocus correction data, and for lenses that support it requests the human readable lens name with 0x82.

Then the camera enters the same initialization loop from the original protocol, using 0x91 instead of 0x90 for lenses with image stabilization, initializes aperture, autofocus counters, and other parameters.

Finally, the lens is either put to sleep, or enters normal polling depending on the camera state.

Normal Idle Loop

While the autofocus and other functions are idle, the camera and lens fall into a basic status polling loop. The loop appears to be

- Check autofocus move counter
- Check AV/AF status
- Check autofocus move counter again
- Get current focal length
- Get scaling and correction values
- Check autofocus move counter
- Check AV/AF status
- Check autofocus move counter again
- Check AV/AF status
- Get Aperture Status/characteristics
- Request base sensitivity
- Repeat

The newer generation(s) follow a very similar loop, with the additional available commands such as the 0xC2, 0xC4, and assorted 0xBn commands.

Focus Adjustment Loop

When focus is requested, which can be the user actively engaging autofocus, or when the camera is set to constantly attempt to retain focus while on, the idle loop is modified.

Following the scaling and correction section the camera gets the updated 0xC0 counter and 0x90/91 status, then issues a move command. It then jumps back into the loop as before, monitoring the 0xC0 counter to track move progress. Additional move commands are issued as required.

Capturing Photos

This section to be completed, all captures not yet acquired.

Summary Of Commands

This is a summary of all known commands, in order. The next section will shed more detail on how they are used and typical command sequences.

Data provided for each command is as follows:

0x<Command Hex Value> - <Confidence that all information is known and correct>

Description	Type	Generation	Length
Command description	Type of Command	Oldest generation this command has been found in	Length in number of Bytes of the sequence including the command byte

(Longer description of the command, data encoded in the command and replies, and other general information)

(Expected reply details from the lens)

	Frame 1	Frame 2	...	Frame N
Camera	Command	Command byte 1	...	Command Byte N
Lens	--	Reply Byte 1	...	Reply Byte N

(where applicable, a summary of known replies or command values)

Confidence Level	Meaning
Complete	Byte is understood completely, no remaining open questions
High	Command structure is understood and most to all decoding has been completed, very few open questions
Medium	Command is roughly understood, but there are enough open questions to consider some parts unknown
Low	Almost nothing is understood about the command, all theories are essentially guesses

Command Type	Meaning
Command - Static	Static command where no additional information is provided by the camera. Typically 1 line commands like initialization commands.
Command + Value	Commands where the camera also provides a value to indicate some information about the expected reaction from the lens. Example: Motor Move Command + Move size.
Command + Values	Same as above, but with either variable length or multiple values being sent by the camera
Request - Static	Request for information from the lens. Camera sends the command and then pads the Camera to Lens data line with filler values (typically 0x00) to allow the lens to reply with the expected number of Bytes.
Request + Value(s)	Request for information where the reply from the lens will change based on the information sent by the camera. Example: the request for lens information/feature negotiation that relies on the camera communicating capabilities to the lens.

Generation – This indicates the earliest generation the command has been observed in.

Length – The number of bytes that make up the command frame. The length is the same for both the camera and lens sides of the bus, however because SPI offsets the camera reply by one frame, the lens side of the bus operates at frame+1 counting. A command listed as a length of 2 will have exactly 2 bytes on each data line.

0x00 - Complete

Description	Type	Generation	Length
Null	Command - Static	1	1

Not really a command. Used as a buffer when waiting for a reply or data from the lens

0x05 – Medium

Description	Type	Generation	Length
Move To Infinity End Stop	Command - Static	1	1

Commands the AF motor to move to infinity end stop. Usually preceded by the 0x0C command. Sometimes never used until the lens advertises status as 0x10 (at end stop) at least once.

Lens must reply with the same value

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x05	--		
Lens	--	0x05		

0x06 – Medium

Description	Type	Generation	Length
Move To Near End Stop	Command - Static	1	1

Commands the AF motor to move to near end stop. Usually preceded by the 0x0C command. Sometimes never used until the lens advertises status as 0x10 (at end stop) at least once.

Lens must reply with the same value

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x06	--		
Lens	--	0x06		

0x07 – Low

Description	Type	Generation	Length
Initialize AF?	Command - Static	2	1

Lesser used command that occurs during the depth of field test with some cameras/lenses. Could be an initialize command.

Lens must reply with the same value

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x07	--	--	--
Lens	--	0x07	--	--

0x08 – Medium

Description	Type	Generation	Length
Put lens to sleep	Command - Static	1	2

Command to put lens into sleep mode. When successful the lens no longer responds to commands until the 0x0A, 0x00 wake sequence is seen. All replies will appear as 0xFF since signal idles high.

Lens returns two bytes

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x08	0x00	--	--
Lens	--	Byte 1	Byte 2	--

Observed replies:

- 0xFF, 0x00: unknown meaning
- 0x08, 0x00: Order refused or not recognized
- 0xFF, 0xAA: Order acknowledged, lens sleeping

0x09 – Medium

Description	Type	Generation	Length
Wake Lens	Command - Static	1	1

Seen when the lens does not reply to the 0x0A command, usually followed by 0x0f and 0x0E. Sometimes seen when lens is already awake, more study is needed

Lens returns one byte, 0x09, when the lens is already in available state.

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x09	--	--	--
Lens	--	0x09	--	--

0x0A – Medium

Description	Type	Generation	Length
Lens Availability Query	Request - Static	1	1

Used to check the status of the lens. This command is usually sent before other sequences of operations, and is always followed with 0x00 to allow the lens to reply before issuing a new command. May always send at the lower 80kHz rate?

Lens returns one byte

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x0A	--	--	--
Lens	--	Byte 1	--	--

Observed replies:

- 0xAA: Acknowledge, lens available.
- 0x0n: Error number (error values unknown)

0x0B – Medium

Description	Type	Generation	Length
Aperture Initialization	Command - Static	1	1

Initializes the aperture motor. Resets bit 1 of Byte 2 of the AF-AV status request command to zero.

Lens returns one byte

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x0B	--	--	--
Lens	--	Byte 1	--	--

Observed replies:

- 0x0B: Returned if successful

0x0C – High

Description	Type	Generation	Length
Autofocus Initialization/reset	Command - Static	1	1

Initializes or resets the AF system in the lens. Resets the AF position counter available through the 0xC0 command to zero. This is used both when the camera turns on to initialize the lens counters, and at the completion of focus sequences to reset the move counter.

Lens returns one byte

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x0C	--	--	--
Lens	--	Byte 1	--	--

Observed replies:

- 0x0C: returned if successful

0x0D – Medium

Description	Type	Generation	Length
End of Shot?	Command - Static	1	1

Appears to be an end of photo notification, sent after every shutter release cycle. Not all camera bodies use it however. It must be replied to correctly, camera will error out if not.

Lens returns one byte, must echo the command

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x0D	--	--	--
Lens	--	0x0D	--	--

0x0E – Low

Description	Type	Generation	Length
Initialization??	Command - Static	1	1

Unknown Command. Appears to be an initialization command, but also appears with 0x0D following the capture of an image.

Lens returns one byte – Appears that the command is expected to be echo'd back.

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x0E	--	--	--
Lens	--	0x0E	--	--

0x0F – Low

Description	Type	Generation	Length
Initialization??	Command - Static	1	1

Unknown Command

Lens returns one byte – Appears that the command is expected to be echo'd back.

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x0FE	--	--	--
Lens	--	0x0F	--	--

0x12 – High

Description	Type	Generation	Length
Aperture Move Command	Command + Value	1	2

Command to open or close the aperture. The value sent indicates the movement size in 1/8 of an aperture increments. Commands are relative to the current position, not an absolute position to move to.

The value is a signed byte that uses the range of 0x01 through 0x7F for positive commands (close aperture to a smaller opening), and 0xFF through 0x80 for negative values (open aperture to larger opening).

Example: For a lens with a base aperture of F/1.8, to close to F/9 the value sent will be 0x25

See the later section about Aperture notation for more information on how step sizes are expressed.

Lens returns two bytes – Should echo the command back for each

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x12	Value (int8_t)	--	--
Lens	--	0x12	0x12	--

0x13 – Medium

Description	Type	Generation	Length
Aperture Move Command	Command + Value	2	2

Nearly Identical to the 0x12 command, but used in newer versions of the protocol.

Expected reply appears to vary, possibly with specific protocol generation. JP's documentation says the reply should be 0x13 0x13, however testing has shown that doing so can throw an Err 01 on newer cameras, and instead the reply needs to be 0x13 0x00 or potentially a full echo of the commanded value.

Lens returns two bytes –

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x13	Value (int8_t)	--	--
Lens	--	0x13	Byte2	--

Empirical observations indicate that the second reply should be 0x00, not 0x13. Replying 0x13 for both sometimes causes Err 01. More investigation needed.

0x30 – Medium

Description	Type	Generation	Length
Modern Focus Move Command	Command + Values	3	4

Appears to be the newer method for commanding focus change, observed in newer lenses such as the 70-300usm ii.

The first two command values function identically to the 0x44 focus move command where the two bytes form a 16bit word representing a signed value indicating the target number of move steps from the current motor position. It is theorized that the third value is a speed command, but this is not confirmed. Also unconfirmed if there are always 3 values or if it can vary. This command appears to always be followed by the 0xFB command. More data is required.

Lens returns 0x30 to all bytes.

	Frame 1	Frame 2	Frame 3	Frame 4	Frame 5
Camera	0x30	Value 1	Value 2	Value 3	--
Lens	--	0x30	0x30	0x30	0x30

0x32 – low

Description	Type	Generation	Length
Unknown	Command - Static	3	1

Unknown command that appears in in newer generations of the protocol. Often follows the 0x33 command, but not always

Lens returns two bytes

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x32	0x00	--	--
Lens	--	Byte 1	Byte2	--

0x33 – Low

Description	Type	Generation	Length
Unknown	Command - Static	3	1

Unknown command that appears in in newer generations of the protocol.

Lens returns four bytes

	Frame 1	Frame 2	Frame 3	Frame 4	
Camera	0x33	0x00 or 0x01	0x00	0x00	
Lens	--	Byte 1	Byte 2	Byte 3	Byte 4

0x44 – Complete

Description	Type	Generation	Length
Focus Motor Move Command	Command + values	1	3

Command to move focus motor by a number of steps sent as a 2's complement signed 16bit integer. The commanded step size is based on the most recently reported value for 0xC0. For example, if 0xC0 is reporting 0, and the command is a value of 100, the lens targets 100 motor steps in the positive direction. If part of the way through this move the lens has reported progress through 0xC0 as 80, and the camera sends -20, the lens will now target a value of 60 by going backwards.

Motor displacement value can vary between -32768 (0x8000) and 32767 (0x7FFF). Negative values are calculated by subtracting from 0x1000, for example $|0xFFFC| = 0x1000 - 0xFFFC = 0x0004$ (two's complement).

Lens returns three bytes

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x44	Value 1	Value 2	--
Lens	--	0x44	Byte1	Byte 2

Some lenses reply with 0x44 for all replies, some echo the values back. It's unclear why there is a difference with some lenses, may be a protocol generation split.

0x50 – Medium

Description	Type	Generation	Length
Set AF Speed	Command + Value(s)	1	2-3

Changes the focus motor speed for lenses that support it. For cases where a single value is sent, there appear to be 4 settings for speed, commanded with the values 0x0C (slowest) through 0x0F (fastest). Range for the dual value option is unknown.

Lens returns two or three bytes

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x50	Value 1	Value 2 (optional)	--
Lens	--	0x50	0x50	0x50 (optional)

0x68 – Low

Description	Type	Generation	Length
Unknown	Command - Static	3	2

Unknown command that appears when autofocus is on. A reply is expected, but value seems to vary lens to lens.

Lens returns two bytes

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x68	0x00	--	--
Lens	--	Byte 1	0x00	--

0x80 – Medium

Description	Type	Generation	Length
Request for characteristics	Request + values	1	8+

Request for information about a lens, used when the lens is attached or when the camera turns on or comes out of sleep. Second Byte seems to always 0x0A. The camera may then send up to 6 bytes depending on the camera and lens. The most basic version sends only 1 value. When 3 values are sent, three frames of 0x00 are then sent to allow the lens to fully answer. If the additional 3 values are sent there is only a single closing 0x00 afterwards

Lens returns up to twelve bytes. Optional values and reply bytes highlighted in orange

	Frame 1	Frame 2	Frame 3	Frame 4	Frame 5	Frame 6	Frame 7
Camera	0x80	0x0A	Value 1	Value 2	Value 3	0x00	0x00
Lens	--	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6

Frame 8	Frame 9	Frame 10	Frame 11	Frame 12	Frame 13	Frame 14
0x00	0x00	Value 4	Value 5	Value 6	0x00	--
Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	0x00

Meaning of the replies is only partially understood.

Reply Byte	Information	Example
1	Lens type (prime, zoom, etc)	See Later Table for values
2	Lens ID Number	See later Table For Values
3 + 4	Minimum Focus Length	0x0046 = 50mm
5 +6	Maximum Focal Length	0x0190 = 400mm
7	Features/protocol version	
8	Features/protocol version	
9	Unknown	
10	Unknown	
11	Unknown	
12	Unknown	

Bytes 7 and 8 relay information at a bit level. Not all items are known

Byte 7

Bit	Feature
7	
6	
5	Lens Has Image stabilization
4	
3	
2	Lens has focus distance encoder
1	
0	Lens can use the C2 command and knows focus distance

Byte 8

Bit	Feature
7	Lens can new newer gen Protocol (faster clock speed, additional commands)
6	
5	
4	
3	
2	Indicated for super telephoto lenses with Memory and AF button
1	
0	Indicated for super telephoto lenses with Memory and AF button

0x81 – Low

Description	Type	Generation	Length
Unknown	Command - Static	2	4

Unknown command appearing in newer generations of the protocol, might be a continuation of the 0x80 command when switching to the faster protocol speed.

Lens returns four bytes

	Frame 1	Frame 2	Frame 3	Frame 4	Frame 5	Frame 6
Camera	0x81	0x13	0x00	0x00	0x00	--
Lens	--	Byte 1	Byte 2	Byte 3	Byte 4	0x00

0x82 – High

Description	Type	Generation	Length
Request for Lens Name	Request - Static	2	variable

Command that requests the human readable name of the lens. Camera sends 0x83 as the follow up command to request the next character until the lens sends a trailing 0x00. Data is encoded as ascii characters.

Lens returns a variable number of Bytes

	Frame 1	Frame 2	Frame 3	Frame 4	...	Frame N	Frame N+1
Camera	0x82	0x83	0x83	0x83	...	0x83	
Lens	--	Byte 1	Byte 2	Byte 3	...	0x00	0x00

Example:

	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6
Lens Hex	0x45	0x46	0x33	0x35	0x6D	0x6S
Lens ASCII	E	F	3	5	M	M

0x83 – High

Not a true standalone command. This is used to request the next character in the 0x82 command when the lens is sending its human readable name.

0x84 – Low

Description	Type	Generation	Length
Unknown Lens Information Request	Request - Static	2	12

Unknown command requesting lens information. Returns 12 bytes depending on the lens and associated characteristics.

Lens returns twelve bytes

	Frame 1	Frame 2	Frame 3	Frame 4	...	Frame 13
Camera	0x84	0x00	0x00	0x00	...	
Lens	--	Byte 1	Byte 2	Byte 3	...	Byte 12

0x86 – Low

Description	Type	Generation	Length
Unknown	Request - Static	2	2

Unknown command requesting lens characteristics.

Lens returns two bytes

	Frame 1	Frame 2	Frame 3
Camera	0x86	0x00	--
Lens	--	Byte 1	Byte 2

0x87 – Low

Description	Type	Generation	Length
Unknown	Request - Static	2	3

Unknown command requesting lens characteristics.

Lens returns two bytes

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x87	0x00	0x00	--
Lens	--	Byte 1	Byte 2	Byte 3

0x90 – Medium

Description	Type	Generation	Length
Autofocus and Aperture System Status	Request - Static	1	3

Request for information about the autofocus and aperture system. The first reply Byte is normally 0x00, The second reply byte encodes the actual status, or returns 0x00 for a system at rest.

Lens returns three bytes.

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0x90	0x00	0x00	--
Lens	--	Byte 1	Byte 2	Byte 3

Meaning of the replies is only partially understood.

Byte 2 relays information at a bit level. Not all items are known

Bit	Feature
7	Lens AF switch is in Manual Mode, or AF button pressed on Super Telephoto Lens
6	
5	AF System error or busy
4	AF initialized and in MFD or Infinity Focus Position
3	
2	AF Motor Actively Moving
1	Aperture out of base position and counter initialized from base position
0	Aperture out of Base position

0x91 – Medium

Description	Type	Generation	Length
Autofocus, Aperture, and Stabilization System Status	Request – Static/value	1	4

Request for information about the autofocus, aperture, and Stabilization system. The follow up value byte is 0x8x for lenses without stabilization, and 0xAx for those with it. Reply is similar to 0x90, but with stabilization data in reply Byte 3 if present, 0x00 if not.

Lens returns four bytes.

	Frame 1	Frame 2	Frame 3	Frame 4	Frame 5	Frame 6
Camera	0x90	Value	0x00	0x00	0x00	--
Lens	--	Byte 1	Byte 2	Byte 3	Byte 4	0x00

Meaning of the replies is only partially understood.

Byte 2 relays information at a bit level. Not all items are known

Bit	Feature
7	Lens AF switch is in Manual Mode, or AF button pressed on Super Telephoto Lens
6	
5	AF System error or busy
4	AF initialized and in MFD or Infinity Focus Position
3	
2	AF Motor Actively Moving
1	Aperture out of base position and counter initialized from base position
0	Aperture out of Base position

Byte 3 relays information at a bit level. Not all items are known

Bit	Feature
7	
6	
5	
4	Indicates something about the system, possibly just a system ready?
3	Stabilization system is in operation
2	Stabilization switch is turned on
1	Stabilization is operational and actively stabilizing
0	

0x93 – Low

Description	Type	Generation	Length
Stabilization Parameter	Command + Value(s)	2	4

Appears to be a command sending characteristics to configure the stabilization, possibly telling it about the in camera stabilization for cameras that have it.

Lens echoes 0x93 to all frames

	Frame 1	Frame 2	Frame 3	Frame 4	Frame 5
Camera	0x93	Value 1	Value 2	Value 3	--
Lens	--	0x93	0x93	0x93	0x93

0x94 – Low

Description	Type	Generation	Length
Electronic Focus Ring Data	Request - Static	2	1

Found on lenses with electronic focus rings. Returns a single byte that changes as the ring is moved. Appears that the information returned depends on the value sent with the command? More investigation required.

Lens returns one byte

	Frame 1	Frame 2	Frame 3
Camera	0x94	--	--
Lens	--	Byte 1	--

0xA0 – Complete

Description	Type	Generation	Length
Request current focal length	Request - Static	1	2

Request for the current focal length of the lens, expressed in mm by combining both values into a 16bit Byte

Lens returns two bytes

	Frame 1	Frame 2	Frame 3
Camera	0xA0	0x00	--
Lens	--	Byte 1	Byte 2

Example: Byte 1 = 0x01 Byte 2 = 0x90, combines to 0x0190 = 400mm

0xB0 – Complete

Description	Type	Generation	Length
Request for Aperture Characteristics	Request - Static	1	3

Request for information about the lens's aperture range. Sends current, max, and min values in Canon's notation (see later section).

Lens returns three bytes.

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0xB0	0x00	0x00	--
Lens	--	Byte 1	Byte 2	Byte 3

Bytes send current, max, and min values

Byte	Value
1	Current aperture opening of the lens
2	Maximum Aperture of the lens (largest opening)
3	Minimum Aperture of the lens (smallest opening)

0xB1 – Low

Description	Type	Generation	Length
Request for Lens information	Request - Static	2	2

Command requesting some lens characteristic, meaning remains unknown at this time

Lens returns two bytes

	Frame 1	Frame 2	Frame 3
Camera	0xB1	0x00	--
Lens	--	Byte 1	Byte 2

0xB2 – Medium

Description	Type	Generation	Length
Unknown AF Correction Parameter	Request - Static	1	3

Command present from the earliest generation through the newest, this appears to be some kind of correction information for the autofocus system, however it's meaning is not yet understood. It is theorized that it is related to aperture based on the 0xB0 and 0xB4 commands being aperture related. More investigation is required to fully decode.

Lens returns three bytes.

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0xB2	0x00	0x00	--
Lens	--	Byte 1	Byte 2	Byte 3

0xB3 – Low

Description	Type	Generation	Length
Request for Lens information	Request - Static	3	3

Command requesting some lens characteristic, meaning remains unknown at this time

Lens returns three bytes

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0xB3	0x00	0x00	--
Lens	--	Byte 1	Byte 2	Byte 3

0xB4 – Medium

Description	Type	Generation	Length
Aperture information in new generation	Request - Static	2	3

This appears to be identical to the 0xB0 command and is used in newer generations of the protocol.

Lens returns three bytes

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0xB4	0x00	0x00	--
Lens	--	Byte 1	Byte 2	Byte 3

0xB6 – Low

Description	Type	Generation	Length
Request for Lens information	Request - Static	3	3

Command requesting some lens characteristic, meaning remains unknown at this time

Lens returns three bytes

	Frame 1	Frame 2	Frame 3	Frame 4
Camera	0xB6	0x00	0x00	--
Lens	--	Byte 1	Byte 2	Byte 3

0xC0 – Complete

Description	Type	Generation	Length
Request for Focus Motor Move Progress	Request - Static	1	2

Request for progress of the focus motor to complete the requested move. Always starts from zero and then tracks relative movement through a complete focus sequence. Value is sent as a 2's complement signed 16bit integer. Movements towards infinity increase the counter while movements away from infinity decrease it. Counter is reset by the 0x0C command at the end of focus sequences

The total range available varies from lens to lens, and is captured in the 0xE4 command.

Lens returns two bytes

	Frame 1	Frame 2	Frame 3
Camera	0xC0	0x00	--
Lens	--	Byte 1	Byte 2

Example: Byte 1 = 0x00, Byte 2 = 0xE6, combines to 0x00E6 = 230 steps moved towards infinity since the focus sequence started

0xC2 – Complete

Description	Type	Generation	Length
Request for Actual Focus Distance	Request - Static	2	4

Request for the actual focus distance in cm for cameras that contain a focus distance encoder. Reply is in 4 bytes that form two 16bit numbers. These two values send the upper and lower distance of the current encoder value. The focus distance encoders have very rough resolution, so they can only return a range. The camera uses this range in combination with the focus motor encoder value to interpolate a more specific distance to put in the image exif data.

Lens returns four bytes

	Frame 1	Frame 2	Frame 3	Frame 4	Frame 5
Camera	0xC2	0x00	0x00	0x00	--
Lens	--	Byte 1	Byte 2	Byte 3	Byte 4

Example:

Byte 1 + Byte 2	Upper focus distance	0x01 0x5E = 0x015E	350cm
Byte 3 + Byte 4	Lower Focus distance	0x00 0xC7 = 0x00C7	199cm

0xC3 – Low

Description	Type	Generation	Length
Request for AF characteristic	Single	3	1

Seems to be additional telemetry on the newest generation of the protocol. The returned value seems linked to the Byte 0 value of the 0xB2 command.

Lens returns one byte

	Frame 1	Frame 2
Camera	0xC3	--
Lens	--	Byte 1

0xC4 – High

Description	Type	Generation	Length
Focus Distance Encoder Index	Request - Static	2	2

This parameter returns a zone count in opposite directions, Byte 1 counts up while Byte 2 counts down. This seems to provide the camera with the current zone, along with an indication of how many zones to each end stop.

Lens returns two bytes

	Frame 1	Frame 2	Frame 3
Camera	0xC4	0x00	--
Lens	--	Byte 1	Byte 2

0xD0, 0xD1, 0xD2, 0xD3, 0xD4 – Low

Description	Type	Generation	Length
Request for all AF characteristics	Request - Static	2 or 3	6 or 10

Request for all values returned by the 0xB0, 0xC0, 0xE0, 0xE4, 0xE8, 0xEA, 0xF8, and 0xFE commands. Appears to be a circular buffer of all replies. Camera sends 0xDF to request next value. 0xD4 access a buffer with only 6 bytes while the rest return 10. Differences need to be studied more to understand what information is omitted with 0xD4.

Lens returns as many bytes as is asked for in a circular buffer

	Frame 1	Frame 2	Frame 3	...	Frame N	Frame N+1	Frame N+2	...
Camera	0xDx	0xDF	0xDF	...	0xDF	0xDF	0xDF	...
Lens	--	Byte 1	Byte 2	...	Byte max	Byte 1	Byte 2	...

0xDF – High

Not a true standalone command. This is used to request the next character in the 0xDx parameter circular buffer.

0xE0 – Medium

Description	Type	Generation	Length
Raw Sensitivity	Request - Static	1	2

Returns Raw/local sensitivity for the lens' optical design, which is used in conjunction with 0xE4 to calculate a required focus motor move. The encoding appears to be

$$E0_{decimal} = 16384 + 1024 * \log_2(Sensitivity)$$

This appears to combine with E4 as

$$Effective \frac{\mu m}{step} = \frac{2^{\left(\frac{E4+E0-163}{1024}\right)}}{64}$$

This explanation again falls apart with the generation 3 lenses as values with the MSB set break the model and require reflecting the value to fit the observed camera calculations. More investigating is required to determine if there is a true shift in generation 3 encoding, or if the proposed encoding is wrong.

The offset is also suspicious and presents more as a way to make the explanation fit the data rather than a clear explanation for why the data is encoded that way. More investigation is required.

Lens returns two bytes

	Frame 1	Frame 2	Frame 3
Camera	0xE0	0x00	--
Lens	--	Byte 1	Byte 2

Example, a reply of 0x40, 0x00 = 0x4000 = 16384

$$16384 = 16384 + 1024 * \log_2(1)$$

Example 2, a reply of 0x3F, 0x83 = 0x3F83 = 16259

$$16256 = 16384 + 1024 * \log_2(0.919)$$

0xE4 – Medium

Description	Type	Generation	Length
Raw um/step for the lens	Request - Static	1	2

Returns the raw um/step movement of the focus unit based on the total travel length and total step range for the lens. Value is log encoded. The formula appears to be:

$$\frac{um}{step} = \frac{2^{E4/1024}}{64}$$

The problem is that newer generations of lenses will use replies that set the MSB, and those values do not work with this encoding. It's unclear if the encoding changed in later generations, or if this is an indication that the proposed encoding is simply not correct. More information is provided in later sections.

Lens returns two bytes

	Frame 1	Frame 2	Frame 3
Camera	0xE4	0x00	--
Lens	--	Byte 1	Byte 2

Example, a reply of 0x24, 0x7A = 0x247A = 9338

$$\frac{2^{(9338/1024)}}{64} = 8.688 \text{ um/step}$$

0xE8/0xEA – Medium

Description	Type	Generation	Length
Local Sensitivity Correction	Request - Static	1	6

Provides local/raw sensitivity fine corrections. Each appears to dominate corrections in a specific focus move direction with 0xE8 dominating moves away from infinity, and 0xEA dominating moves towards infinity. While there appear to be three 16bit words, each encoded in a Q6.10 log format, word 3 appears as 0x0000 for every lens that data has been collected from.

Values appear to be at least loosely tied to the 0xC4 bins, however it has been observed that the 0xE8 and 0xEA replies do change inside 0xC4 bins on some lenses.

The exact encoding and usage of the replies is still being investigated, however two general trends look promising. The first is that (E8 + EA)/2 approximates the local correction value,

and E8-EA approximates the sensitivity curvature across the range of the lens. More investigation is required to nail down the exact encoding and meaning.

Lens returns six bytes

	Frame 1	Frame 2	Frame 3	Frame 4	Frame 5	Frame 6	Frame 7
Camera	0xE8/EA	0x00	0x00	0x00	0x00	0x00	--
Lens	--	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6

0xEE – Medium

Description	Type	Generation	Length
Request for teleconverter serial number	Request - Static	2	6

On cameras that support it, this requests the serial number of the teleconverter when one is detected. The first returned byte indicates the type of teleconverter, 0xEE for versions before vIII, 0x02 for a vIII 2x converter, and possibly 0x01 for a 1.4x vIII converter. I do not have any teleconverters and cannot prove the values. Bytes 2 through 5 return the serial number in 4 bit bytes, ie a value of 0x27 directly converts to the decimal values 2 and 7.

Lens returns six bytes

	Frame 1	Frame 2	Frame 3	Frame 4	Frame 5	Frame 6	Frame 7
Camera	0xEE	0x00	0x00	0x00	0x00	0x00	--
Lens	--	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6

0xF0 – Low

Description	Type	Generation	Length
Autofocus correction	Request - Static	old	1

Unknown command. In the older generation it is always sent immediately after the 0xB2 request, which suggests that it might be some kind of factory calibration correction value. More data is required.

Lens returns one byte

	Frame 1	Frame 2
Camera	0xF0	--
Lens	--	Byte 1

0xF8 – Low

Description	Type	Generation	Length
Request for AF characteristic	Request - Static	3	1

Unknown command requesting what appears to be autofocus correction values. Reply seems to be tied to the C4 value or some other lookup table.

Lens returns one byte

	Frame 1	Frame 2
Camera	0xF8	--
Lens	--	Byte 1

0xFA – Low

Description	Type	Generation	Length
Request for AF characteristic	Request - Static	3	1

Unknown command requesting what appears to be autofocus correction values. Reply seems to be tied to the C4 value or some other lookup table.

Lens returns one byte

	Frame 1	Frame 2
Camera	0xFA	--
Lens	--	Byte 1

0xFB – Low

Description	Type	Generation	Length
Unknown	Potentially Command + Value	3	2

Unknown command, always sent immediately after the 0x30 motor move command.

Unsure if it is a standalone command or part of 0x30, but looks like its own command. All the examples I've seen so far send 0x19 as the following value.

Lens returns two bytes, only examples I have echo 0xFB to both command and value

	Frame 1	Frame 2	Frame 3
Camera	0xFB	Value	--
Lens	--	0xFB	???

0xFC – Low

Description	Type	Generation	Length
Request for AF characteristic	Request - Static	3	1

Unknown command requesting what appears to be autofocus correction values. Reply seems to be tied to the C4 value or some other lookup table.

Lens returns one byte

	Frame 1	Frame 2
Camera	0xFC	--
Lens	--	Byte 1

0xFE – Low

Description	Type	Generation	Length
Request for AF characteristic	Request - Static	3	1

Unknown command requesting what appears to be autofocus correction values. Reply seems to be tied to the C4 value or some other lookup table.

Lens returns one byte

	Frame 1	Frame 2
Camera	0xFE	--
Lens	--	Byte 1

0xFD – Low

Description	Type	Generation	Length
Unknown	Request - Static	3	1

Unknown command. Empirically the reply only seems meaningful when the lens in question is a USM lens, but this is not proven.

Lens returns one byte

	Frame 1	Frame 2
Camera	0xFD	--
Lens	--	Byte 1

Summary of Commands available in each Protocol Generation

The following table (Still being compiled and completed) summarizes which generation each command is actively used in.

DRAFT

Commands in More Detail

This section covers specific commands in more detail than the summary section, and includes data used to derive the explanations.

0x80 – Request for Lens Information

This is the initial handshake between the camera and lens. This command is sent both when the lens is first connected, and each time the camera turns on or wakes from sleep. This command serves to allow the camera to know what generation of protocol to use, and informs what features the lens has.

The 0x80 byte appears to always be followed by 0x0A. Depending on the camera being used there can be additional values sent from the camera that potentially could be sending information about the camera. The following bytes have been observed:

Camera	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6
EOS M6 mkii	0x80	0x0A	0xA8	0x37	0x00	0x7F
EOS M50 mkii	0x80	0x0A	0xA8	0x37	0x00	0x7F
EOS RP	0x80	0x0A	0xBF	0xB7	0x01	0x7F
EOS 500	0x80	0x0A	0x00	0x00	0x00	
EOS 5000	0x80	0x0A	0x03	0x00	0x00	
EOS 30D	0x80	0x0A	0x97	0x01	0x00	
EOS 650D	0x80	0x0A	0x99	0x01	0x00	
EOS 40D	0x80	0x0A	0x99	0x01	0x00	
EOS 7D	0x80	0x0A	0x9F	0x03	0x00	
EOS 5D mkiii	0x80	0x0A	0xA8	0x07	0x01	

Items highlighted in Green have been Personally verified, items highlighted in blue are taken from JP's findings (JP, Canon EF - EOS serial protocol, 2016).

The information returned by the lens is as follows:

Reply Byte	Information	Example
1	Lens type (prime, zoom, etc)	See Later Table for values
2	Lens ID Number	See later Table For Values
3 + 4	Minimum Focus Length	0x0046 = 50mm
5 +6	Maximum Focal Length	0x0190 = 400mm
7	Features/protocol version	
8	Features/protocol version	
9	Unknown	
10	Unknown	
11	Unknown	
12	Unknown	

Bytes 7 and 8 are the most interesting as they communicate actual features.

A summary of seen replies and implied feature meaning is still being completed and will be inserted to future revisions.

Physical Lens Focus Status – 0xC0, 0xC2, 0xC3, 0xC4

These four commands return information about the physical status of the lens in terms of focus.

0xC0 – Autofocus movement counter

Present on all generations of the EF protocol, this is how the lens communicates progress on achieving the requested autofocus moves from the lens. This is a relative counter that is reset to zero following every focusing sequence.

The value returned to the camera is a 2's complement signed 16bit number, counting from zero, with moves towards infinity increasing the count, and moves away from infinity decreasing it. The counter has a theoretical decimal range of -32767 to 32767.

The counter appears to be tied to different physical things in different lenses. With the EF 50mm f/1.8 II it appears to be a direct count of the encoder on the focus motor. The following chart shows a timeline of an actual focus sequence. The camera was pointed at a target that was closer than the point previously focused on, and then data collected while the autofocus was commanded on. The MotorA and MotorB signals are the actual terminals on the DC focus motor, used to confirm the motor drive direction. Encoder is the raw encoder signal, while encoderEdges is a running count of the total edges seen.

Throughout the focus sequence the lens is reporting a C0 value in line with the counted encoder edges. The overshoot seen on the first and second moves is the lens adding additional moves to account for backlash in the system. The move immediately before this sequence was in the opposite direction, so the lens added roughly 13 extra edge counts to the first requested move command and did not add them to the C0 count. Same on the second move that changed directions. The following two smaller moves track almost exactly between raw edges and C0.

Noise on the encoder lines generated some extra counts in my capture, these were cleaned up as best I could, but it's possible a handful of spurious counts exist in the Encoder data. Still, the capture shows enough real data to be confident that this lens ties 0xC0 to the motor encoder.

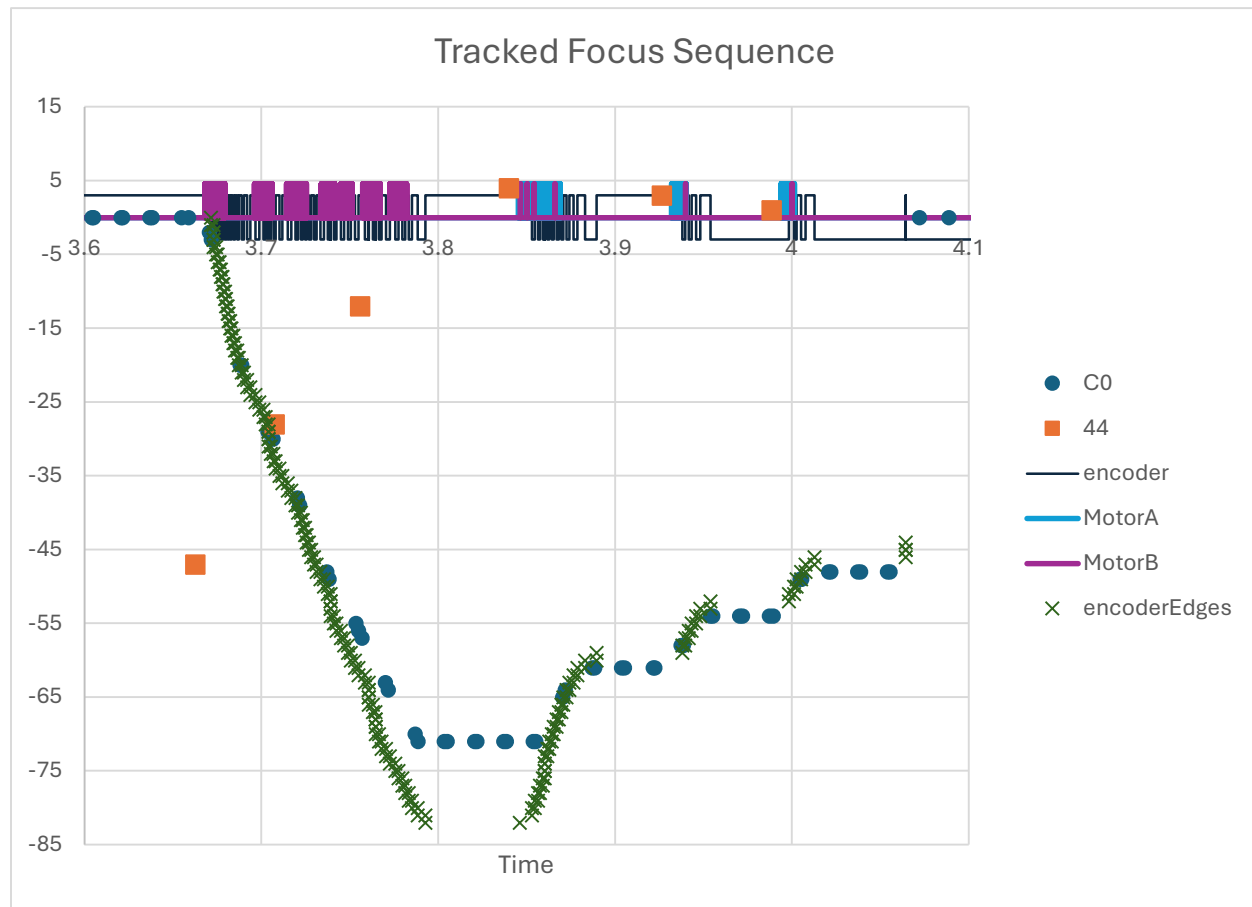


Figure 2: Tracked Focus Chart

On newer lenses the C0 counter appears to be tied to other physical items. For example, in the 50mm f/1.8 STM the gear train is characterized to be 542:1 (motor turns to focus ring turns), and with a calculated $\sim 43^\circ$ of turn on the internal focus ring, that means the motor turns ~ 64.7 times. This lens has a 5 blade encoder wheel with two photocells which should give 20 pulses per rotation for a total of ~ 1290 . However the total C0 range has been observed to be in the range of 4200-4300.

This implies that the C0 count is tied to something else, and the strongest theory is that it is raw microsteps of the motor. Stepper motors of the size used very commonly use a step

angle of 18° for 20 steps per rotation. It is also common to use microstepping to achieve more granular control, and micro step sizes are commonly 2, 4, 8, etc. Using 4 microsteps would give a total range of ~5160 microsteps. The difference between this theorized value and the observed C0 range can be explained by the error in my calculation of the total focus ring motion, combined with the observed behavior that the lens does not seem to run the focus mechanism into the hard stops at each end during normal operation.

A teardown and investigation of a USM lens is needed to look at what might be used in that technology.

One note about newer lenses. In several captures from the 70-300 USM III it has been observed that the C0 counter alters with no apparent move command issued. The working theory is that in more modern lenses the IS system can adjust focus distance as part of the stabilization. Most of the data collected up to this point has purposely been with older lenses lacking IS where this effect would not show up. Further investigation will be performed, and I am especially curious to see if this behavior is communicated in the IS status polls.

0xC2 – Focus Distance

Introduced in the second generation of the EF protocol this parameter reports a focus distance range to the camera. The resolution is low due to the very coarse encoder used in the lens. The mapping of encoder position to distance is done by the lens, and the upper/lower distance for each position is reported to the camera. The camera body then interpolates the specific focus distance based on the C0 value to embed a focus distance in the EXIF data.

The following chart shows the decimal value of the captured 0xC2 values from the 50mm f/1.8 STM during a manual focus sweep from infinity to min focus distance and back to infinity. The reported C2 values change as the C4 bin value changes.

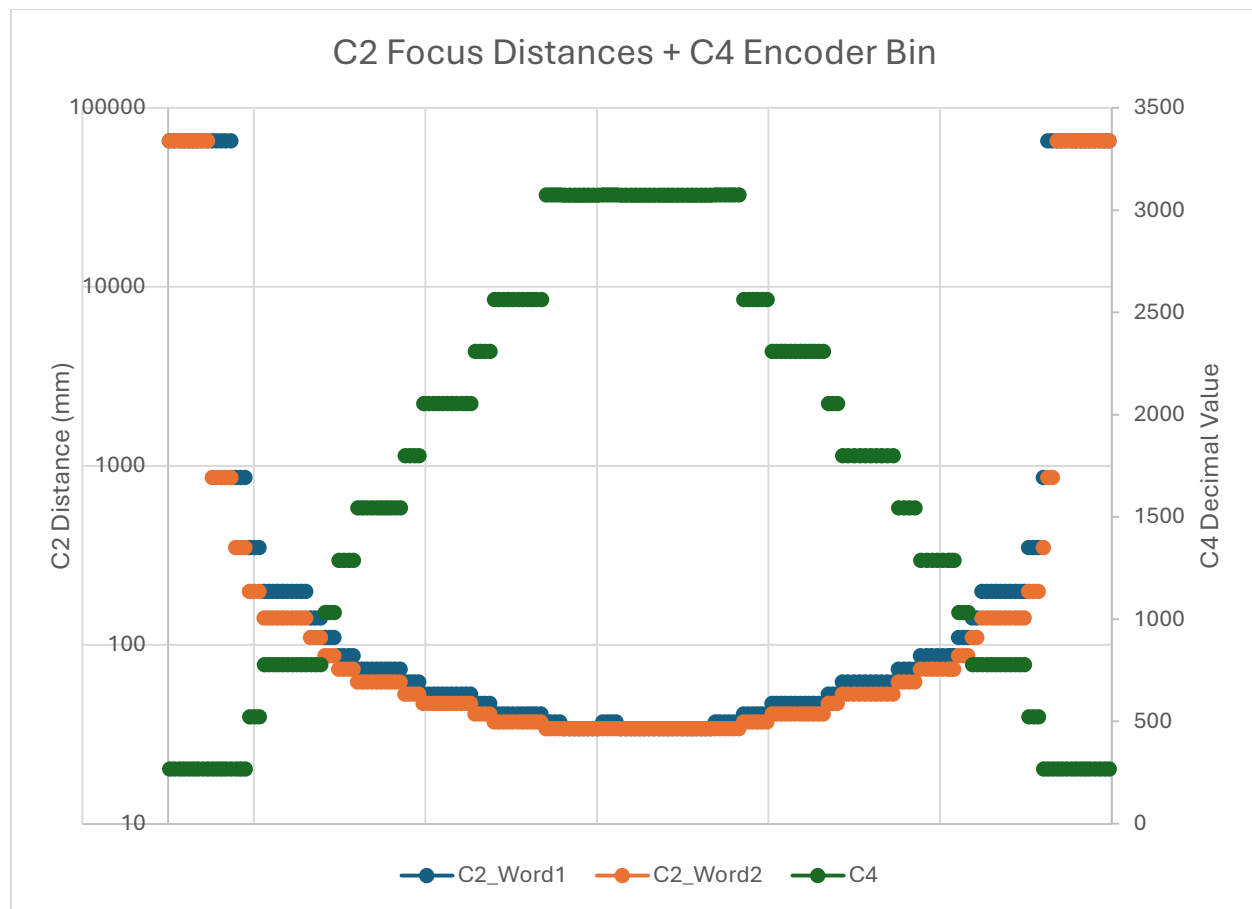


Figure 3: Focus Distance and Encoder Bin Overlay

One interesting detail about the EF-s 18-55 f/3.5-5.6 IS STM specifically, the lens does not contain a physical focus distance encoder, however it still reports C4 bins and ties other replies to them. This implies that it is tracking based on step count and could report a far more granular focus distance, but was specifically configured not to.

0xC3 – Unknown Telemetry

Seen only on the newest lenses, this value seems to track directly with byte 0 from the 0xB2 command. More investigation is required to characterize this parameter.

0xC4 – Raw Encoder Position

Introduced in the second generation of the EF protocol, this single parameter is perhaps one of the most important to be added. The value reported is the actual encoder position read from the course focus distance encoder in the lens. The values that are reported for all the new autofocus correction values seem largely binned to the C4 position.

The values of the two bytes appear to be counting in opposite directions to provide a constant view of the current zone and distance to each end stop. Example values from the EF 50mm f/1.8 STM

C4 Hex	Whole Word Decimal	Byte 1 Hex	Byte 2 Hex	Byte 1 Decimal	Byte 2 Decimal
010C	268	01	0C	1	12
010B	267	01	0B	1	11
020B	523	02	0B	2	11
030A	778	03	0A	3	10
0409	1033	04	09	4	9
0508	1288	05	08	5	8
0608	1544	06	08	6	8
0707	1799	07	07	7	7
0806	2054	08	06	8	6
0905	2309	09	05	9	5
0A03	2563	0A	03	10	3
0C02	3074	0C	02	12	2
0C01	3073	0C	01	12	1

The following chart shows just the reported C4 values from the EF 50mm f/1.8 STM across the full focus range. Though hard to see, the difference between the Whole word decimal and the Word 1 decimal appears to be the Word 2 decimal. The trend is clear however that the two words count opposite each other.

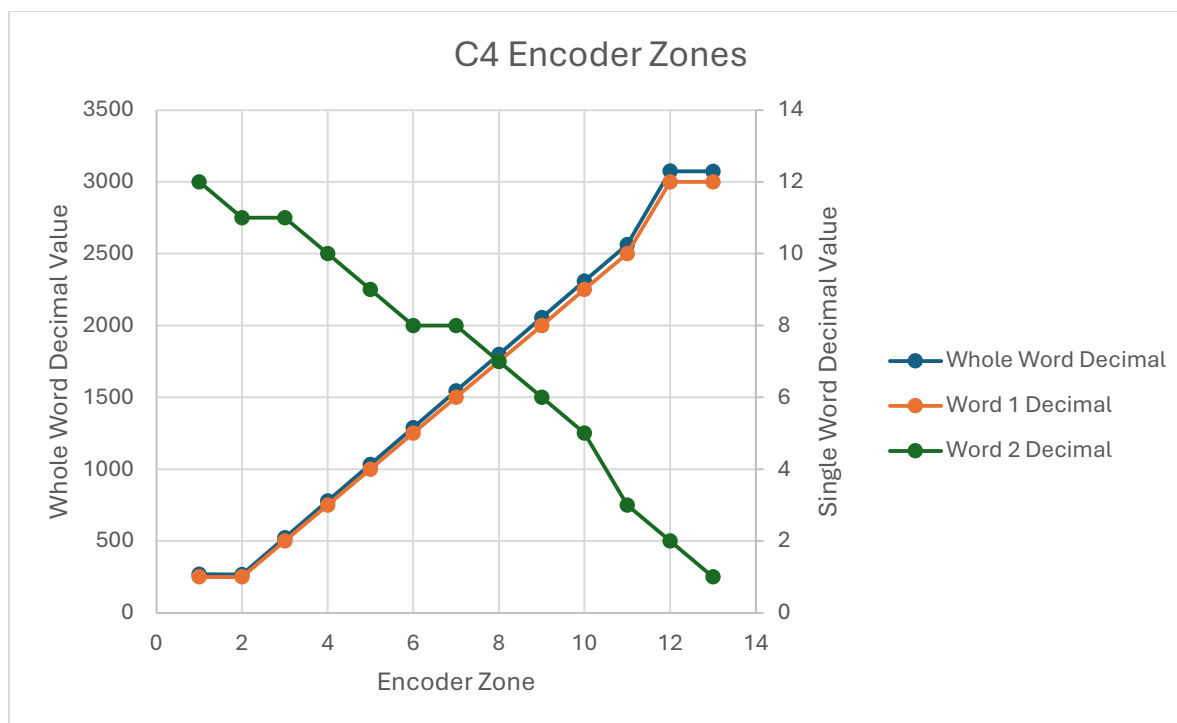


Figure 4: C4 Encoder Zones

An open question remains on why sometimes one word will change before the other. Additionally on zoom lenses it is common for the range presented to change based on the zoom distance. This makes sense for lenses that use specific parts of the total focus unit travel for different zoom distances, however more lenses need to be studied to fully understand how the zones are defined, especially on newer STM lenses that do not have a physical encoder and appear to be deriving one based on the stepper range.

Core Auto Focus Parameters – 0xE0, 0xE4, 0xE8, 0xEA

Present from the earliest lenses through to the most recent, these parameters provide the core scaling that the camera uses to convert a measured defocus value to a commanded motor move size. They are always polled in the same order of 0xE0, 0xEA, 0xE8, 0xE4, perhaps indicating the order that they are applied during the camera's calculations.

0xE0 – Local Raw sensitivity

This parameter appears to carry the raw sensitivity value of the lens based on zoom and focus distance. Combined with the E4 $\mu\text{m}/\text{step}$ value this would allow the camera to calculate a required focus move size needed to correct the measured defocus at the sensor.

On the EF 50mm f/1.8 ii, the reported E0 value is always 0x4000. This lens also has a sensitivity that is effectively 1 across the whole range. Since it is a unit focus lens (all elements move together), sensitivity is calculated as

$S = \frac{d\delta}{dx} = 1 - \beta^2$ where δ is the defocus distance at the sensor and β is the magnification of the moving group.

β is always 0 at infinity, and for this lens it is 0.15x at the near end, which gives a sensitivity range of 1 to 0.978. On a lens this early without any focus distance encoders it would make sense to simply report a sensitivity of 1 and allow the control loop to work out whatever small error is introduced.

This would indicate that the value of 0x4000 is being used as sensitivity of 1 and that the general encoding is

$$\text{Sensitivity} = 2^{(E0-16 / 1024)}$$

Testing this theory, and knowing that the value should double or halve with a decimal change of 1024, that should mean that 0x4400 is a sensitivity of 2, and 0x3C00 should be a sensitivity of 0.5. Testing this theory by overriding E0 with those values, I do see the commanded move sizes half and double for each tested defocus.

Looking now at the EF-s 18-55 STM, this lens is an internal group focus rather than a unit group. This means that sensitivity is not constant and does in fact vary with zoom distance. If we assume for a moment that the 16384 offset is real, this lens reports E0 values ranging from 0x4BB9 at 55mm to 0x3F9E at 18mm. This would give sensitivity values of 7.62 through 0.935, which should be correct for this lens.

Again there is a problem with generation 3 lenses. Looking at the 50mm f/1.8 STM, this lens should also have a sensitivity that is roughly 1 as its magnification at the near end is 0.21, which gives $1 - 0.1156 = 0.956$. Over the full focus range we see E0 span 0xCE5E to 0xCEA4, which completely break the fit. This would suggest that either generation 3 is encoding differently, or that my encoding model is incorrect.

Connecting the 50mm STM to an older 20D, it still reports the same values, implying that generation 2 understands how to work with the MSB encoding. More investigation will be required.

0xE8/0xEA – Focus Direction Specific Scaling

These two parameters appear to work in tandem to provide sensitivity corrections based on the direction of focus movement. 0xE8 dominates moves away from infinity, while 0xEA dominates moves towards infinity. The exact decoding has not been fully completed, however overriding values of E8 and EA shows a measurable change in the commanded move sizes.

More data is needed to fully characterize the impact from these parameters.

0xE4 – Global scaling/raw $\mu\text{m}/\text{step}$

This parameter provides the raw $\mu\text{m}/\text{step}$ when calculated with the total focus unit travel and total number of steps. This value is log encoded, presumably to make math operations easier on the earliest generation microcontrollers.

Using the man in the middle testing device many values of E4 were tested with the EF 50mm f/1.8 ii to study the impact on the commanded focus behavior. In each test a target was moved to a known position and focus was established with the camera. Then the target was moved a known amount and the focus was again acquired while overriding the 0xE4 reply. This allows for studying the change in commanded move behavior. Below is a chart showing the first commanded move size at several defocus sizes compared to the used E4 value. While there is a lot of noise in the individual points, the trend is fairly clear. As the E4 increases, the commanded size shrinks. Additionally, the size of the commanded move doubles/halves at an interval of 1024 units of E4. This supports the idea that E4 encodes the raw $\mu\text{m}/\text{step}$ of the focus drive motor, as the $\mu\text{m}/\text{step}$ gets larger you need fewer steps to move to the desired position.

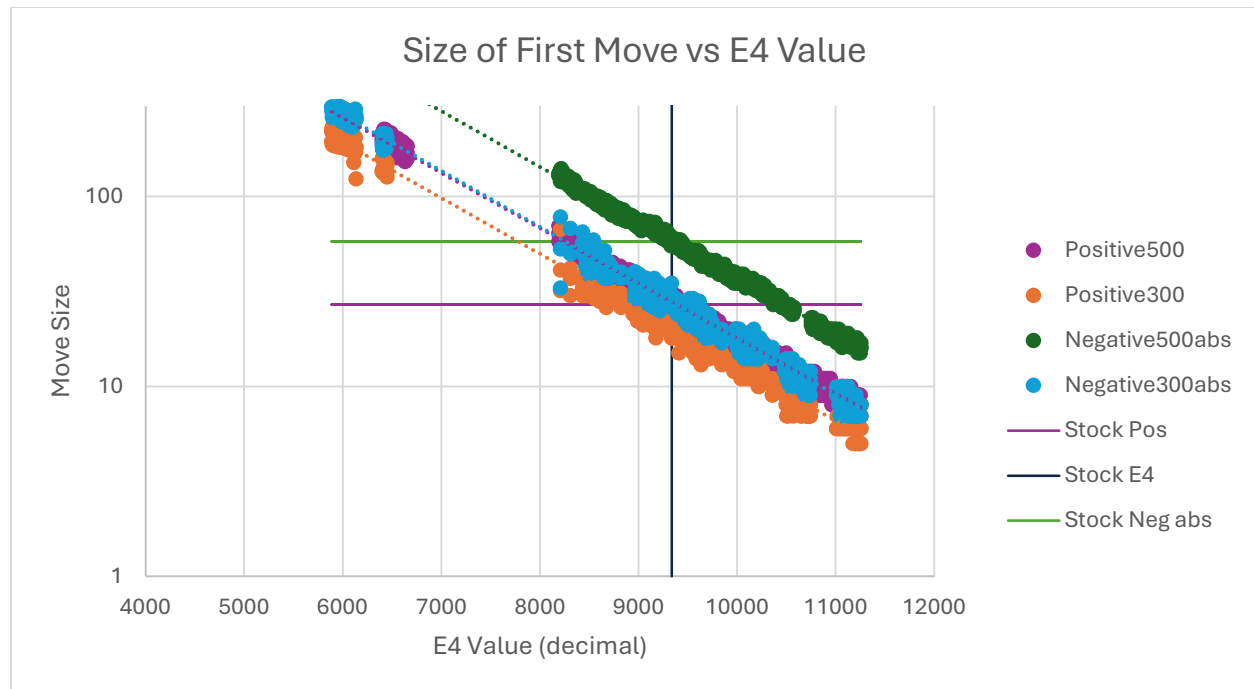


Figure 5: First Move Size vs E4 Value

The final model fit equation works out to

$$\frac{um}{step} = \frac{2^{(E4/1024)}}{64}$$

There is an open question related to the generation 2 and 3 implementation, no values that set the MSB work with this model. Take for example the 50mm f/1.8 STM, which uses a reply of 0x93E7, which is 37863 decimal. Used directly this would give

$$\frac{2^{(37863/1024)}}{64} = \frac{2^{36.97}}{64} = 2111448536 \mu m/step$$

It's tempting to see this as simply a new scaling since 2.1 $\mu m/step$ is a plausible value for this lens, however more proof is required to determine if this is real or a coincidence. Several similar coincidences led to false explanations for the E4 encoding up to this point.

There is also a problem with how generation 1 control loops react to values that set the MSB. If generation 1 simply ignored the MSB, you would expect to see identical commands when using a reply of 0x4000 and 0xC000, however this is not the case. Using 0xC000 results in commanded move sizes that more closely align with those seen with 0x4400. It is unclear at this time if this indicates that generation 1 never expected to see that MSB set, or

if this is an indication that my proposed encoding simply is incomplete or incorrect. This pattern repeats itself with other parameters as well.

Focus Motor Moves – 0x44, 0x30

In practice, requesting focus moves happens the same way in all generations of the protocol. The camera requests the current C0 value as the final step in calculating the required move, then issues a move as a number of steps to move. The camera is able to issue new commands while the focus motor is running in order to update the target end position.

The move size is sent as a 2's complement signed 16bit number and always acts as a modification to the most recent C0 value, not a modification to the previously sent target.

For example, if the counter is at zero, and the camera issues a request to move to 500, then halfway through it reads the C0 at 200 and issues a new command of 250, the new target is 450, not 750.

0x30 – New Move Command

The new 0x30 command used in newer generations of lenses adds a third command Byte. The function of this byte is still being studied, with the working theory that it may be an integrated speed command. The 0x50 Set AF speed command seems to only ever appear once at initialization for new lenses and never again, so this may be a merge of the two in order to be able to quickly ask for varying speed moves.

The command is almost always followed immediately by the 0xFB command. Originally I thought this was some kind of “end of frame” marker for the new move command, but it seems to be its own command and needs further investigation.

Canon Aperture Notation

Aperture values are converted to a decimal value for the sake of use in the EF protocol. In JP's document the term AV-EF is created to refer to these values (JP, Canon EF - EOS serial protocol, 2016), I will continue to use that name here as well.

The values used in the EF protocol are an expression of aperture in 1/8 stops. Not all lenses can move in 1/8 stops, commonly using 1/4 stop intervals. Even more confusingly, many cameras use 1/3 stop intervals, so it is not uncommon for the setting on the camera to be slightly different from the actual setting in the lens.

For following terms are used in the conversions and tables

N: The common f number represented as either f/N or 1:N

AV-EF: The value sent by the camera or lens as a Byte

AV: Aperture expressed in APEX notation

Conversion	Equation
$N > AV-EF$	$AVEF = (16 * \log_2(N)) + 8$
$AV-EF > N$	$N = 2^{\left(\frac{AVEF-8}{16}\right)}$
$AV > AV-EF$	$AVEF = (AV + 1) * 8$
$AV-EF > AV$	$AV = \left(\frac{AVEF}{8}\right) - 1$
$N > AV$	$AV = 2 * \log_2(N)$
$AV > N$	$N = \sqrt{2^{AV}}$

The following table provides pre-calculated conversions between all notations

N (1/3)	1	1.1		1.3	1.4		1.6		1.8	2			2.5
N (1/4)	1	1.1	1.2	1.3	1.4	1.5		1.7	1.8	2	2.2	2.4	
AV	0	0.28	0.53	0.76	0.97	1.17	1.36	1.53	1.7	2	2.28	2.53	2.64
AVEF	8	10	12	14	16	17	19	20	22	24	26	28	29

N (1/3)		2.8		3.2		3.5		4		4.5		5	
N (1/4)	2.6	2.8	3.1		3.4		3.7	4	4.4		4.8		5.2
AV	2.76	2.97	3.26	3.36	3.53	3.61	3.78	4	4.28	4.34	4.53	4.64	4.76
AVEF	30	32	34	35	36	37	38	40	42	43	44	45	46

N (1/3)	5.6		6.3		7		8		9		10		11
N (1/4)	5.6	6.2		6.7		7.3	8	8.7		9.5		10.4	
AV	4.97	5.26	5.31	5.49	5.61	5.74	6	6.24	6.34	6.5	6.64	6.76	6.92
AVEF	48	50	50	52	53	54	56	58	59	60	61	62	63

N (1/3)			12.7		14.3		16		18		20		22
N (1/4)	11.3	12.3		13.5		14.7	16	17.4		19		20.7	
AV	7	7.24	7.33	7.51	7.68	7.76	8	8.24	8.34	8.5	8.64	8.74	8.92
AVEF	64	66	67	68	69	70	72	74	75	76	77	78	79

N (1/3)			25		28		32						
N (1/4)	22.6	24.7		26.9		29.3	32	34.9	38.1	41.5	45.3	49.4	53.8
AV	9	9.25	9.29	9.5	9.61	9.75	10	10.25	10.5	10.75	11	11.25	11.5
AVEF	80	82	82	84	85	86	88	90	92	94	96	98	100

Lens ID Numbers

The following tables are copied from JP's document (JP, Canon EF - EOS serial protocol, 2016) and can be verified through several other sources online. Third party IDs have been moved to a separate Table.

At some point Canon expanded the lensID field to use two bytes as the original 255 values were all used. Only one byte seems to be asked for, and lenses using a 2 byte reply will answer with the second byte in the initial 0x80 request for information.

Lens ID	Hex	Model
1	01	EF 50mm f/1.8
2	02	EF 28mm f/2.8
3	03	EF 135mm f/2.8 Soft
4	04	EF 35-105mm f/3.5-4.5 or Sigma Lens
5	05	EF 35-70mm f/3.5-4.5
6	06	EF 28-70mm f/3.5-4.5 or Sigma or Tokina Lens
7	07	EF 100-300mm f/5.6L
8	08	EF 100-300mm f/5.6 or Sigma or Tokina Lens
9	09	EF 70-210mm f/4
10	0A	EF 50mm f/2.5 Macro or Sigma Lens
11	0B	EF 35mm f/2
13	0D	EF 15mm f/2.8 Fisheye
14	0E	EF 50-200mm f/3.5-4.5L
15	0F	EF 50-200mm f/3.5-4.5
16	10	EF 35-135mm f/3.5-4.5
17	11	EF 35-70mm f/3.5-4.5A
18	12	EF 28-70mm f/3.5-4.5
20	14	EF 100-200mm f/4.5A
21	15	EF 80-200mm f/2.8L
22	16	EF 20-35mm f/2.8L gold Tokina Lens
23	17	EF 35-105mm f/3.5-4.5
24	18	EF 35-80mm f/4-5.6 Power Zoom
25	19	EF 35-80mm f/4-5.6 Power Zoom
26	1A	EF 100mm f/2.8 Macro or Other Lens
27	1B	EF 35-80mm f/4-5.6
28	1C	EF 80-200mm f/4.5-5.6 gold Tamron Lens
29	1D	EF 50mm f/1.8 II
30	1E	EF 35-105mm f/4.5-5.6
31	1F	EF 75-300mm f/4-5.6 gold Tamron Lens
32	20	EF 24mm f/2.8 or Sigma Lens
35	23	EF 35-80mm f/4-5.6
36	24	EF 38-76mm f/4.5-5.6
37	25	EF 35-80mm f/4-5.6 gold Tamron Lens
38	26	EF 80-200mm f/4.5-5.6
39	27	EF 75-300mm f/4-5.6
40	28	EF 28-80mm f/3.5-5.6
41	29	EF 28-90mm f/4-5.6

42	2A	EF 28-200mm f/3.5-5.6
43	2B	EF 28-105mm f/4-5.6
44	2C	EF 90-300mm f/4.5-5.6
45	2D	EF-S 18-55mm f/3.5-5.6 [II]
46	2E	EF 28-90mm f/4-5.6
48	30	EF-S 18-55mm f/3.5-5.6 IS
49	31	EF-S 55-250mm f/4-5.6 IS
50	32	EF-S 18-200mm f/3.5-5.6 IS
51	33	EF-S 18-135mm f/3.5-5.6 IS
52	34	EF-S 18-55mm f/3.5-5.6 IS II
53	35	EF-S 18-55mm f/3.5-5.6 III
54	36	EF-S 55-250mm f/4-5.6 IS II
94	5E	TS-E 17mm f/4L
95	5F	TS-E 24.0mm f/3.5 L II
124	7C	MP-E 65mm f/2.8 1-5x Macro Photo
125	7D	TS-E 24mm f/3.5L
126	7E	TS-E 45mm f/2.8
127	7F	TS-E 90mm f/2.8
129	81	EF 300mm f/2.8L
130	82	EF 50mm f/1.0L
131	83	EF 28-80mm f/2.8-4L
132	84	EF 1200mm f/5.6L
134	86	EF 600mm f/4L IS
135	87	EF 200mm f/1.8L
136	88	EF 300mm f/2.8L
137	89	EF 85mm f/1.2L
138	8A	EF 28-80mm f/2.8-4L
139	8B	EF 400mm f/2.8L
140	8C	EF 500mm f/4.5L
141	8D	EF 500mm f/4.5L
142	8E	EF 300mm f/2.8L IS
143	8F	EF 500mm f/4L IS
144	90	EF 35-135mm f/4-5.6 USM
145	91	EF 100-300mm f/4.5-5.6 USM
146	92	EF 70-210mm f/3.5-4.5 USM
147	93	EF 35-135mm f/4-5.6 USM
148	94	EF 28-80mm f/3.5-5.6 USM
149	95	EF 100mm f/2 USM
150	96	EF 14mm f/2.8L
151	97	EF 200mm f/2.8L
152	98	EF 300mm f/4L IS
153	99	EF 35-350mm f/3.5-5.6L
154	9A	EF 20mm f/2.8 USM or
155	9B	EF 85mm f/1.8 USM
156	9C	EF 28-105mm f/3.5-4.5 USM
160	A0	EF 20-35mm f/3.5-4.5 USM o
161	A1	EF 28-70mm f/2.8L
162	A2	EF 200mm f/2.8L
163	A3	EF 300mm f/4L

164	A4	EF 400mm f/5.6L
165	A5	EF 70-200mm f/2.8 L
166	A6	EF 70-200mm f/2.8 L + 1.4x
131	83	EF 28-80mm f/2.8-4L
132	84	EF 1200mm f/5.6L
134	86	EF 600mm f/4L IS
135	87	EF 200mm f/1.8L
136	88	EF 300mm f/2.8L
137	89	EF 85mm f/1.2L
138	8A	EF 28-80mm f/2.8-4L
139	8B	EF 400mm f/2.8L
140	8C	EF 500mm f/4.5L
141	8D	EF 500mm f/4.5L
142	8E	EF 300mm f/2.8L IS
143	8F	EF 500mm f/4L IS
144	90	EF 35-135mm f/4-5.6 USM
145	91	EF 100-300mm f/4.5-5.6 USM
146	92	EF 70-210mm f/3.5-4.5 USM
147	93	EF 35-135mm f/4-5.6 USM
148	94	EF 28-80mm f/3.5-5.6 USM
149	95	EF 100mm f/2 USM
150	96	EF 14mm f/2.8L
151	97	EF 200mm f/2.8L
152	98	EF 300mm f/4L IS
153	99	EF 35-350mm f/3.5-5.6L
154	9A	EF 20mm f/2.8 USM
155	9B	EF 85mm f/1.8 USM
156	9C	EF 28-105mm f/3.5-4.5 USM
160	A0	EF 20-35mm f/3.5-4.5 USM
161	A1	EF 28-70mm f/2.8L
162	A2	EF 200mm f/2.8L
163	A3	EF 300mm f/4L
164	A4	EF 400mm f/5.6L
165	A5	EF 70-200mm f/2.8 L
166	A6	EF 70-200mm f/2.8 L + 1.4x
167	A7	EF 70-200mm f/2.8 L + 2x
168	A8	EF 28mm f/1.8 USM
169	A9	EF 17-35mm f/2.8L
170	AA	EF 200mm f/2.8L II
171	AB	EF 300mm f/4L
172	AC	EF 400mm f/5.6L
173	AD	EF 180mm Macro f/3.5L
174	AE	EF 135mm f/2L
175	AF	EF 400mm f/2.8L
176	B0	EF 24-85mm f/3.5-4.5 USM
177	B1	EF 300mm f/4L IS
178	B2	EF 28-135mm f/3.5-5.6 IS
179	B3	EF 24mm f/1.4L
180	B4	EF 35mm f/1.4L

181	B5	EF 100-400mm f/4.5-5.6L IS + 1.4x
182	B6	EF 100-400mm f/4.5-5.6L IS + 2x
183	B7	EF 100-400mm f/4.5-5.6L IS
184	B8	EF 400mm f/2.8L + 2x
185	B9	EF 600mm f/4L IS
186	BA	EF 70-200mm f/4L
187	BB	EF 70-200mm f/4L + 1.4x
188	BC	EF 70-200mm f/4L + 2x
189	BD	EF 70-200mm f/4L + 2.8x
190	BE	EF 100mm f/2.8 Macro USM
191	BF	EF 400mm f/4 DO IS
193	C1	EF 35-80mm f/4-5.6 USM
194	C2	EF 80-200mm f/4.5-5.6 USM
195	C3	EF 35-105mm f/4.5-5.6 USM
196	C4	EF 75-300mm f/4-5.6 USM
197	C5	EF 75-300mm f/4-5.6 IS USM
198	C6	EF 50mm f/1.4 USM
199	C7	EF 28-80mm f/3.5-5.6 USM
200	C8	EF 75-300mm f/4-5.6 USM
201	C9	EF 28-80mm f/3.5-5.6 USM
202	CA	EF 28-80mm f/3.5-5.6 USM IV
208	D0	EF 22-55mm f/4-5.6 USM
209	D1	EF 55-200mm f/4.5-5.6
210	D2	EF 28-90mm f/4-5.6 USM
211	D3	EF 28-200mm f/3.5-5.6 USM
212	D4	EF 28-105mm f/4-5.6 USM
213	D5	EF 90-300mm f/4.5-5.6 USM
214	D6	EF-S 18-55mm f/3.5-5.6 USM
215	D7	EF 55-200mm f/4.5-5.6 II USM
224	E0	EF 70-200mm f/2.8L IS
225	E1	EF 70-200mm f/2.8L IS + 1.4x
226	E2	EF 70-200mm f/2.8L IS + 2x
227	E3	EF 70-200mm f/2.8L IS + 2.8x
228	E4	EF 28-105mm f/3.5-4.5 USM
229	E5	EF 16-35mm f/2.8L
230	E6	EF 24-70mm f/2.8L
231	E7	EF 17-40mm f/4L
232	E8	EF 70-300mm f/4.5-5.6 DO IS USM
233	E9	EF 28-300mm f/3.5-5.6L IS
234	EA	EF-S 17-85mm f/4-5.6 IS USM
235	EB	EF-S 10-22mm f/3.5-4.5 USM
236	EC	EF-S 60mm f/2.8 Macro USM
237	ED	EF 24-105mm f/4L IS
238	EE	EF 70-300mm f/4-5.6 IS USM
239	EF	EF 85mm f/1.2L II
240	F0	EF-S 17-55mm f/2.8 IS USM
241	F1	EF 50mm f/1.2L
242	F2	EF 70-200mm f/4L IS
243	F3	EF 70-200mm f/4L IS + 1.4x

244	F4	EF 70-200mm f/4L IS + 2x
245	F5	EF 70-200mm f/4L IS + 2.8x
246	F6	EF 16-35mm f/2.8L II
247	F7	EF 14mm f/2.8L II USM
248	F8	EF 200mm f/2L IS or Sigma Lens
249	F9	EF 800mm f/5.6L IS
250	FA	EF 24mm f/1.4L II or Sigma Lens
251	FB	EF 70-200mm f/2.8L IS II USM
252	FC	EF 70-200mm f/2.8L IS II USM + 1.4x
253	FD	EF 70-200mm f/2.8L IS II USM + 2x
254	FE	EF 100mm f/2.8L Macro IS USM
488	01E8	EF-S 15-85mm f/3.5-5.6 IS USM
489	01E9	EF 70-300mm f/4-5.6L IS USM
490	01EA	EF 8-15mm f/4L Fisheye USM
491	01EB	EF 300mm f/2.8L IS II USM
492	01EC	EF 400mm f/2.8L IS II USM
493	01ED	EF 500mm f/4L IS II USM or EF 24-105mm f4L IS USM
493	01ED	EF 24-105mm f/4L IS USM
494	01EE	EF 600mm f/4.0L IS II USM
495	01EF	EF 24-70mm f/2.8L II USM
496	01F0	EF 200-400mm f/4L IS USM
499	01F3	EF 200-400mm f/4L IS USM + 1.4x
502	01F6	EF 28mm f/2.8 IS USM
503	01F7	EF 24mm f/2.8 IS USM
504	01F8	EF 24-70mm f/4L IS USM
505	01F9	EF 35mm f/2 IS USM
506	01FA	EF 400mm f/4 DO IS II USM
507	01FB	EF 16-35mm f/4L IS USM
508	01FC	EF 11-24mm f/4L USM
747	02EB	EF 100-400mm f/4.5-5.6L IS II USM
748	02EC	EF 100-400mm f/4.5-5.6L IS II USM + 1.4x
750	02EE	EF 35mm f/1.4L II USM
4142	102E	EF-S 18-135mm f/3.5-5.6 IS STM
4143	102F	EF-M 18-55mm f/3.5-5.6 IS STM
4144	1030	EF 40mm f/2.8 STM
4145	1031	EF-M 22mm f/2 STM
4146	1032	EF-S 18-55mm f/3.5-5.6 IS STM
4147	1033	EF-M 11-22mm f/4-5.6 IS STM
4148	1034	EF-S 55-250mm f/4-5.6 IS STM
4149	1035	EF-M 55-200mm f/4.5-6.3 IS STM
4150	1036	EF-S 10-18mm f/4.5-5.6 IS STM
4152	1038	EF 24-105mm f/3.5-5.6 IS STM
4153	1039	EF-M 15-45mm f/3.5-6.3 IS STM
4154	103A	EF-S 24mm f/2.8 STM
4155	103B	EF-M 28mm f/3.5 Macro IS STM
4156	103C	EF 50mm f/1.8 STM
36912	9030	EF-S 18-135mm f/3.5-5.6 IS USM

Lens ID (copied)	Copy #	Hex	Brand	Model
4	1	04	Sigma	UC Zoom 35-135mm f/4-5.6
6	1	06	Sigma	18-50mm f/3.5-5.6 DC
6	2	06	Sigma	18-125mm f/3.5-5.6 DC IF ASP
6	3	06	Tokina	AF 193-2 19-35mm f/3.5-4.5
6	4	06	Sigma	28-80mm f/3.5-5.6 II Macro
8	1	08	Sigma	70-300mm f/4-5.6 [APO] DG Macro
8	2	08	Tokina	AT-X 242 AF 24-200mm f/3.5-5.6
9	1	09	Sigma	55-200mm f/4-5.6 DC
10	1	0A	Sigma	50mm f/2.8 EX
10	2	0A	Sigma	28mm f/1.8
10	3	0A	Sigma	105mm f/2.8 Macro EX
10	4	0A	Sigma	70mm f/2.8 EX DG Macro EF
22	1	16	Tokina	AT-X 280 AF Pro 28-80mm f/2.8 Aspherical
26	1	1A	Cosina	100mm f/3.5 Macro AF
26	2	1A	Tamron	SP AF 90mm f/2.8 Di Macro
26	3	1A	Tamron	SP AF 180mm f/3.5 Di Macro
26	4	1A	Zeiss	Planar T* 50mm f/1.4
28	1	1C	Tamron	SP AF 28-105mm f/2.8 LD Aspherical IF
28	2	1C	Tamron	SP AF 28-75mm f/2.8 XR Di LD Aspherical [IF] Macro
28	3	1C	Tamron	AF 70-300mm f/4-5.6 Di LD 1:2 Macro
28	4	1C	Tamron	AF Aspherical 28-200mm f/3.8-5.6
31	1	1F	Tamron	SP AF 300mm f/2.8 LD IF
32	1	20	Sigma	15mm f/2.8 EX Fisheye
33	1	21	Voigtlander	Ultron 40mm f/2 SLII Aspherical
33	2	21	Voigtlander	Color Skopar 20mm f/3.5 SLII Aspherical
33	3	21	Voigtlander	APO-Lanthar 90mm f/3.5 SLII Close Focus
33	4	21	Zeiss	Distagon T* 15mm f/2.8 ZE
33	5	21	Zeiss	Distagon T* 18mm f/3.5 ZE
33	6	21	Zeiss	Distagon T* 21mm f/2.8 ZE
33	7	21	Zeiss	Distagon T* 25mm f/2 ZE
33	8	21	Zeiss	Distagon T* 28mm f/2 ZE
33	9	21	Zeiss	Distagon T* 35mm f/2 ZE
33	10	21	Zeiss	Distagon T* 35mm f/1.4 ZE
33	11	21	Zeiss	Planar T* 50mm f/1.4 ZE
33	12	21	Zeiss	Makro-Planar T* 50mm f/2 ZE
33	13	21	Zeiss	Makro-Planar T* 100mm f/2 ZE
33	14	21	Zeiss	Apo-Sonnar T* 135mm f/2 ZE
37	1	25	Tamron	70-200mm f/2.8 Di LD IF Macro
37	2	25	Tamron	AF 28-300mm f/3.5-6.3 XR Di VC LD Aspherical [IF] Macro Model A20
37	3	25	Tamron	SP AF 17-50mm f/2.8 XR Di II VC LD Aspherical [IF]
37	4	25	Tamron	AF 18-270mm f/3.5-6.3 Di II VC LD Aspherical [IF] Macro
43	1	2B	Tamron	AF 28-300mm f/3.5-6.3 XR Di VC LD Aspherical [IF] Macro Model A20
47	1	2F	Zeiss	Milvus 35mm f/2 or 50mm f/2
47	1	2F	Zeiss	Milvus 50mm f/2 Makro
131	1	83	Sigma	8mm f/3.5 EX DG Circular Fisheye
131	2	83	Sigma	17-35mm f/2.8-4 EX DG Aspherical HSM
131	3	83	Sigma	17-70mm f/2.8-4.5 DC Macro
131	4	83	Sigma	APO 50-150mm f/2.8 [II] EX DC HSM
131	5	83	Sigma	APO 120-300mm f/2.8 EX DG HSM
131	6	83	Sigma	4.5mm f/2.8 EX DC HSM Circular Fisheye
131	7	83	Sigma	70-200mm f/2.8 APO EX HSM
137	1	89	Sigma	18-50mm f/2.8-4.5 DC OS HSM
137	2	89	Sigma	50-200mm f/4-5.6 DC OS HSM
137	3	89	Sigma	18-250mm f/3.5-6.3 DC OS HSM
137	4	89	Sigma	24-70mm f/2.8 IF EX DG HSM
137	5	89	Sigma	18-125mm f/3.8-5.6 DC OS HSM
137	6	89	Sigma	17-70mm f/2.8-4 DC Macro OS HSM C
137	7	89	Sigma	17-50mm f/2.8 OS HSM
137	8	89	Sigma	18-200mm f/3.5-6.3 DC OS HSM [II]

137	9	89	Tamron	AF 18-270mm f/3.5-6.3 Di II VC PZD
137	10	89	Sigma	8-16mm f/4.5-5.6 DC HSM
137	11	89	Tamron	SP 17-50mm f/2.8 XR Di II VC
137	12	89	Tamron	SP 60mm f/2 Macro Di II
137	13	89	Sigma	10-20mm f/3.5 EX DC HSM
137	14	89	Tamron	SP 24-70mm f/2.8 Di VC USD
137	15	89	Sigma	18-35mm f/1.8 DC HSM
137	16	89	Sigma	12-24mm f/4.5-5.6 DG HSM II
143	1	8F	Sigma	17-70mm f/2.8-4 DC Macro OS HSM
150	1	96	Sigma	20mm EX f/1.8
150	2	96	Sigma	30mm f/1.4 DC HSM
150	3	96	Sigma	24mm f/1.8 DG Macro EX
150	4	96	Sigma	28mm f/1.8 DG Macro EX
152	1	98	Sigma	12-24mm f/4.5-5.6 EX DG ASPHERICAL HSM
152	2	98	Sigma	14mm f/2.8 EX Aspherical HSM
152	3	98	Sigma	10-20mm f/4-5.6
152	4	98	Sigma	100-300mm f/4
153	1	99	Sigma	50-500mm f/4-6.3 APO HSM EX
153	2	99	Tamron	AF 28-300mm f/3.5-6.3 XR LD Aspherical [IF] Macro
153	3	99	Tamron	AF 18-200mm f/3.5-6.3 XR Di II LD Aspherical [IF] Macro Model A14
153	4	99	Tamron	18-250mm f/3.5-6.3 Di II LD Aspherical [IF] Macro
154	1	9A	Zeiss	Milvus 21mm f/2.8
156	1	9C	Tamron	SP 70-300mm f/4.0-5.6 Di VC USD
156	2	9C	Tamron	SP AF 28-105mm f/2.8 LD Aspherical IF
160	1	A0	Tamron	AF 19-35mm f/3.5-4.5
160	2	A0	Tokina	AT-X 124 AF Pro DX 12-24mm f/4
160	3	A0	Tokina	AT-X 107 AF DX 10-17mm f/3.5-4.5 Fisheye
160	4	A0	Tokina	AT-X 116 AF Pro DX 11-16mm f/2.8
160	5	A0	Tokina	AT-X 11-20 F2.8 PRO DX Aspherical 11-20mm f/2.8
161	1	A1	Sigma	24-70mm f/2.8 EX
161	2	A1	Sigma	28-70mm f/2.8 EX
161	3	A1	Sigma	24-60mm f/2.8 EX DG
161	4	A1	Tamron	AF 17-50mm f/2.8 Di-II LD Aspherical
161	5	A1	Tamron	90mm f/2.8
161	6	A1	Tamron	SP AF 17-35mm f/2.8-4 Di LD Aspherical IF
161	7	A1	Tamron	SP AF 28-75mm f/2.8 XR Di LD Aspherical [IF] Macro
168	1	A8	Sigma	50-100mm f/1.8 DC HSM A
169	1	A9	Sigma	18-200mm f/3.5-6.3 DC OS
169	2	A9	Sigma	15-30mm f/3.5-4.5 EX DG Aspherical
169	3	A9	Sigma	18-50mm f/2.8 Macro
169	4	A9	Sigma	50mm f/1.4 EX DG HSM
169	5	A9	Sigma	85mm f/1.4 EX DG HSM
169	6	A9	Sigma	30mm f/1.4 EX DC HSM
169	7	A9	Sigma	35mm f/1.4 DG HSM
172	1	AC	Sigma	150-600mm f/5-6.3 DG OS HSM S
173	1	AD	Sigma	180mm EX HSM Macro f/3.5
173	2	AD	Sigma	APO Macro 150mm f/2.8 EX DG HSM
174	1	AE	Sigma	70-200mm f/2.8 EX DG APO OS HSM
174	2	AE	Sigma	50-500mm f/4.5-6.3 APO DG OS HSM
174	3	AE	Sigma	150-500mm f/5-6.3 APO DG OS HSM
174	4	AE	Zeiss	Milvus 100mm f/2 Makro
180	1	B4	Sigma	50mm f/1.4 DG HSM A
180	2	B4	Sigma	24mm f/1.4 DG HSM A
180	3	B4	Zeiss	Milvus 50mm f/1.4
180	4	B4	Zeiss	Milvus 85mm f/1.4
180	5	B4	Zeiss	Otus 28mm f/1.4 ZE
181	1	B5	Sigma	150-600mm f/5-6.3 DG OS HSM S+1.4x
182	1	B6	Sigma	150-600mm f/5-6.3 DG OS HSM S+2x
183	1	B7	Sigma	150mm f/2.8 EX DG OS HSM APO Macro
183	2	B7	Sigma	105mm f/2.8 EX DG OS HSM Macro
183	3	B7	Sigma	180mm f/2.8 EX DG OS HSM APO Macro

183	4	B7	Sigma	150-600mm f/5-6.3 DG OS HSM C
183	5	B7	Sigma	150-600mm f/5-6.3 DG OS HSM S

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Testing Hardware

All data captures were made using modified lens mount adapters on both EF-m and RF cameras. See the How Mirrorless Lenses Detect EF Lenses section for more information on how EF lenses are detected.

In the case of the EF to EF-m adapter the stock connection between the camera and lens side pins was severed and wires were soldered to the pins on each side before being routed out of the adapter. This allowed each side to be isolated or joined depending on what testing was being performed.

In the case of the Ef to RF adaptor the stock connection was left in place and wires were only soldered to the lens side pins. This is due to the differences in how the EF pins are connected to the RF pins compared to the EF-m pins. See the Physical Interface and Protocol Detection sections for more information.

Initial data capture was performed using the Saleae Logig 2 software and a basic USB logic analyzer. Because of the acknowledge behavior used by the lens on the clock line, the built in SPI decoder will not function correctly. A custom high level analyzer has since been written for logic2. See Appendix B – Testing Scripts and Programs for details on where to find and use it. Captures shown in this document use this custom analyzer.

Raw captures were initially exported and processed using Python scripts and Excel Macros. See Appendix B – Testing Scripts and Programs for script names and descriptions.

High level schematics for all testing setups are provided in Appendix A – Testing Hardware Schematics.

EF to EF-m Adaptor

For testing with EF-m cameras a Viltrox EF to EF-m adapter was modified. Internally the connection between the camera side contacts and lens side contacts was severed. Wires were connected to the internal side of each contact and then routed out of the adapter. This allows completely severing the lens and camera, or connecting the two sides for pass through.

This adapter was used in the following setups:

1. EF-m Camera, EF lens, passive communication recording, used to capture normal communication for analysis.
2. EF-m Camera, EF Lens, Man in the Middle Attack, Used to alter stock lens replies and observe focus changes from the camera.

3. EF-m Camera, Manual Focus Lens, Used for testing my prototype EF Lens Emulator

The following photos show the wire colors attached to the internal pins of the adapter. Note that the order of the wires/pads on the lens side do not align with the pin order of the actual lens contact pins. Several signals are re-routed on the PCB to put them in the correct order for the EF-m pins.

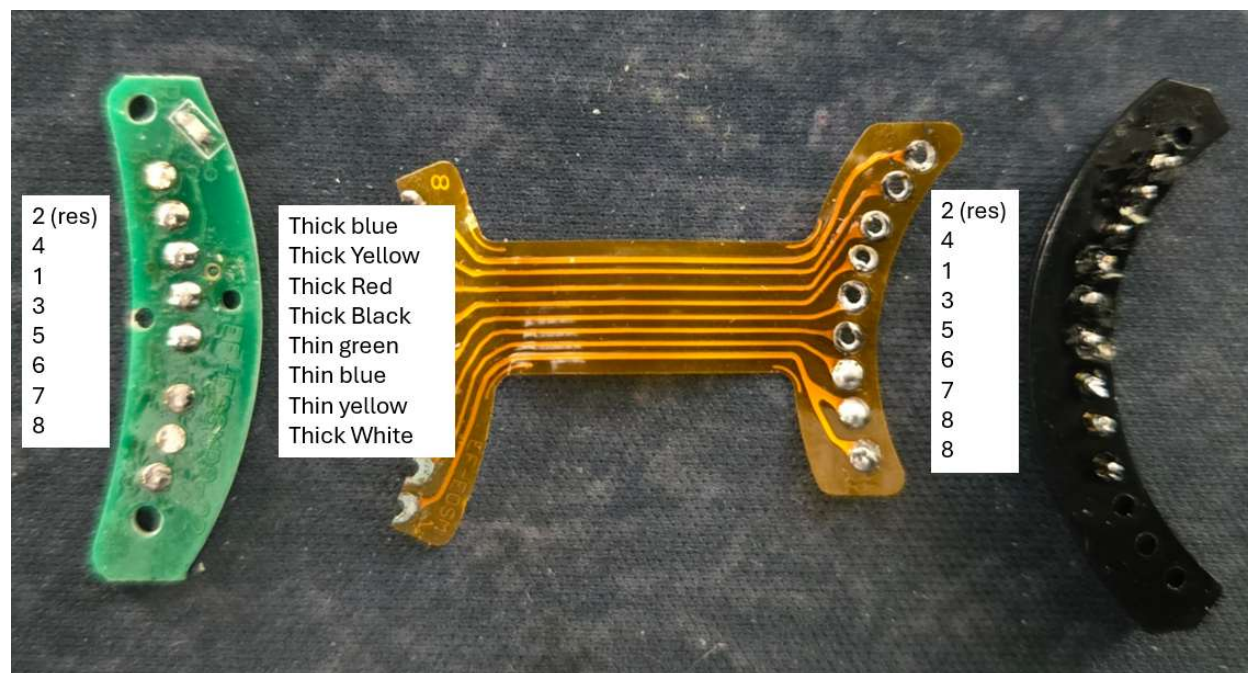


Figure 6: EF to EF-m Adapter Internal Connections

Signal Number	EF Lens signal Name
1	VBAT – Power for assorted lens motors
2	P-GND (shared with 3) – ground for assorted lens motors
3	P-GND (shared with 2) – ground for assorted lens motors
4	VDD – Power for logic circuits
5	Data Camera to Lens
6	Data Lens to Camera
7	Data Clock Line
8	D-GND – Ground for logic circuits

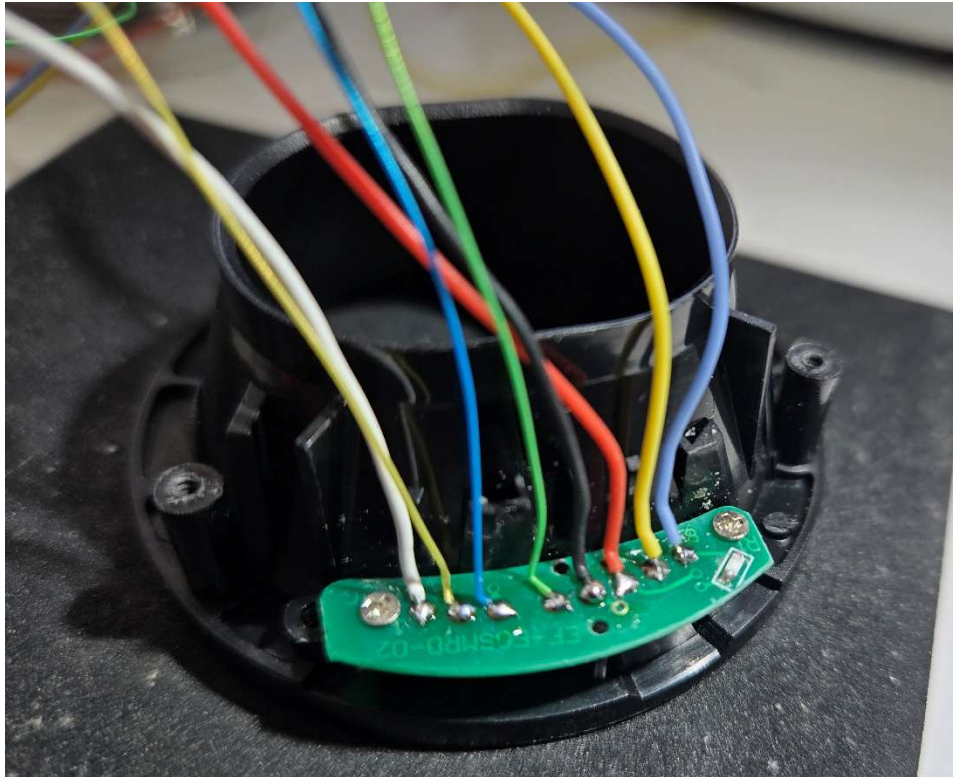


Figure 7: Lens Side Connections

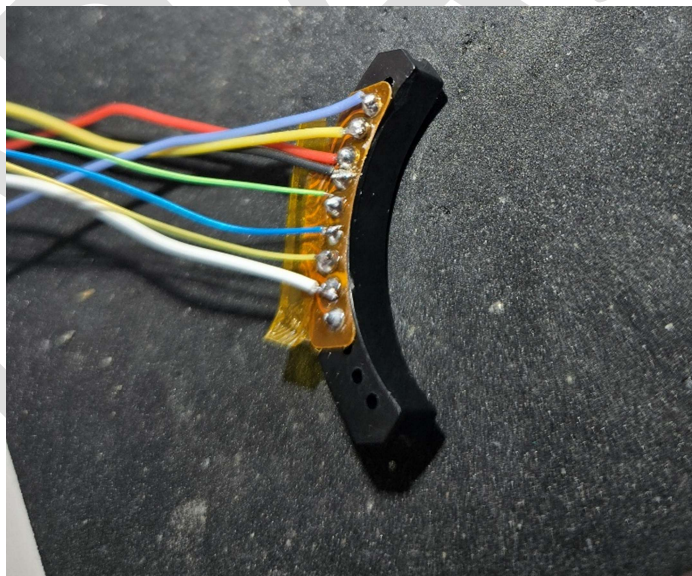


Figure 8: Camera Side Connections

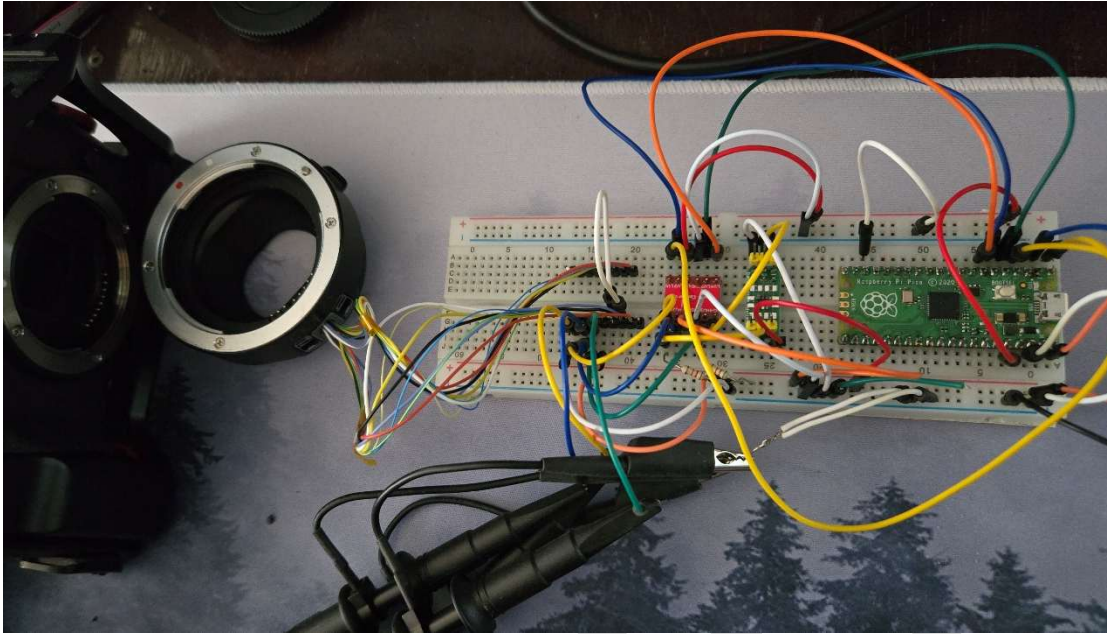


Figure 9: Modified Lens Adapter In Use

EF to RF Adaptor

For testing with RF cameras a MEIKE EF to RF adapter was modified. All internal connections were left in place, instead wires were tapped into the EF side and routed out of the adapter. This allows passive recording, and allows RF cameras to interface with my EF lens emulator, but no Man in the Middle attacks can be performed with this setup.

I unfortunately forgot to take detailed photos of the modifications to this adapter.



Figure 10: EF to RF Modified Adapter

Man in the Middle Testing Setup

This setup was built to facilitate changing very specific lens replies and observing how the camera responded. A microcontroller reads the camera/lens communication and in real time can break the lens to camera data line in order to inject a modified value.

The circuit Schematic for this setup is provided in Appendix A – Testing Hardware Schematics

This setup uses the modified EF to EF-m adapter as the physical interface.

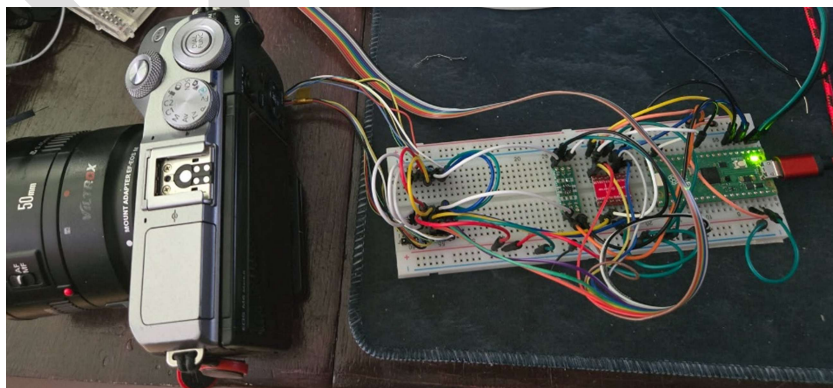


Figure 11: Man in the Middle Testing Setup

Automated Testing Rig

In order to collect data in a controlled way there was a need to be able to repeatedly move a focus target to known distances. A linear track was assembled that allowed a target to be moved from directly in front of the camera to any distance up to 4meters away. Tests could then be carried out overriding certain parameters for known defocus sizes and observing how the camera's focus logic changed. Longer tests could be run unattended in order to gather data from larger sets of override parameters.

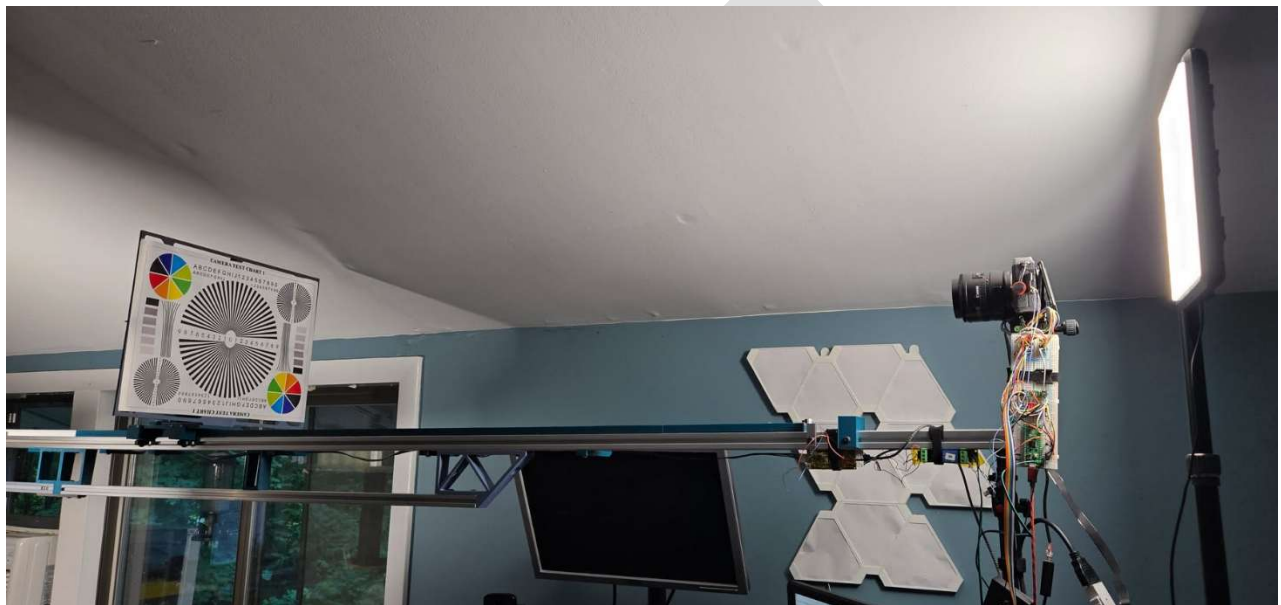


Figure 12: Automated Testing Rig

Works Cited

Franke, J. (2018, July 26). *Canon's EF-M Mount's Lens Type Detection*. Retrieved from Points in Focus: <https://www.pointsinfocus.com/learning/cameras-lenses/canon-ef-m-mount-lens-type-detection/>

JP. (2016, March 11). *Canon EF - EOS serial protocol*. Retrieved from Le portail à JP (Translated: The Gateway to JP): http://jp79dsfr.free.fr/_Docs%20et%20infos/Photo%20Tech%20_%20Canon%20EOS-EF%20Protocol.pdf

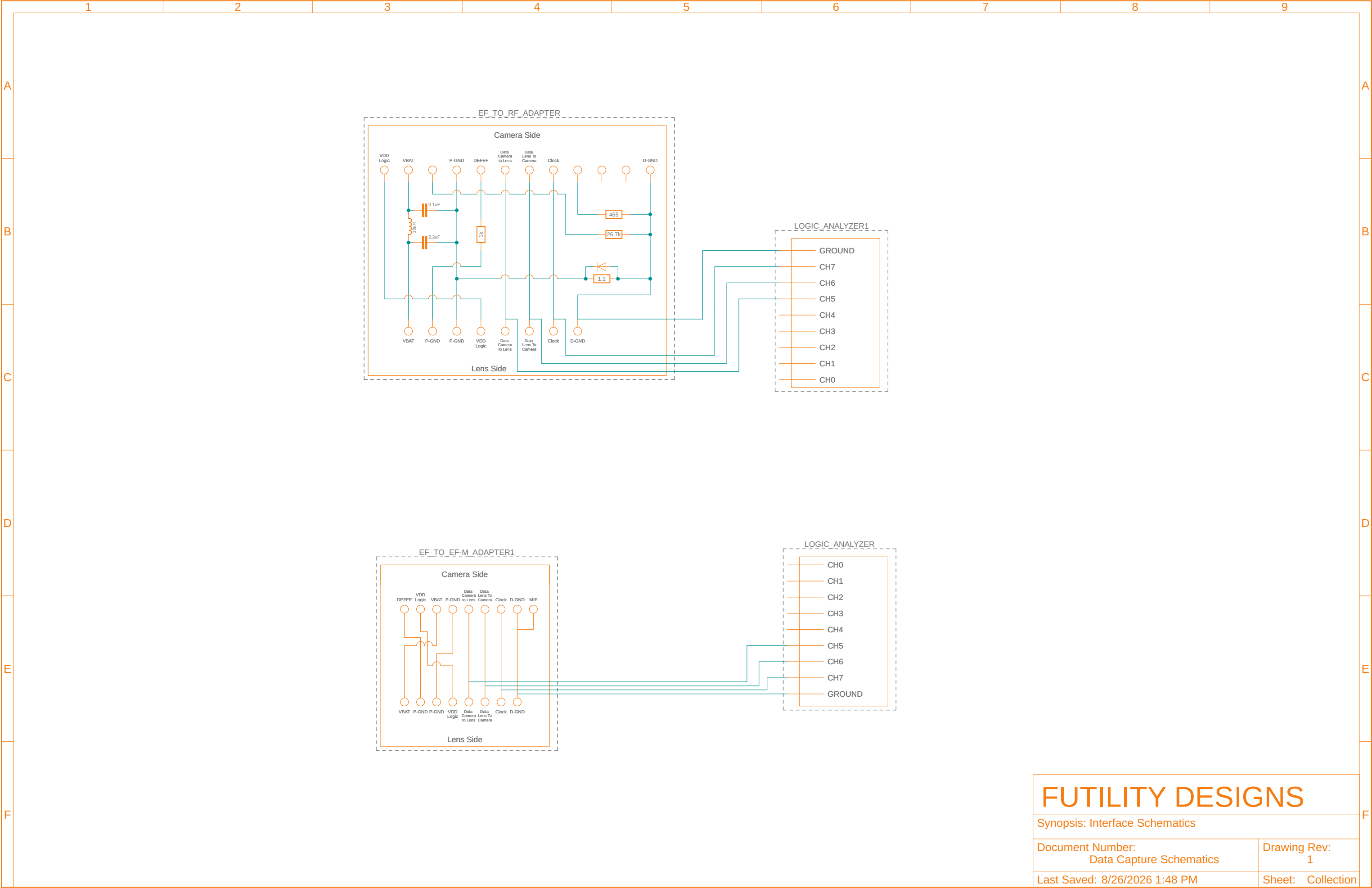
JP. (2016). *PHOTOGRAPHY - TECHNOLOGY: Characteristics and Theory*. Retrieved 2026, from Le portail à JP (Translated: The Gateway to JP): http://jp79dsfr.free.fr/Photo-Tech/Pages/Page_Technologie.htm

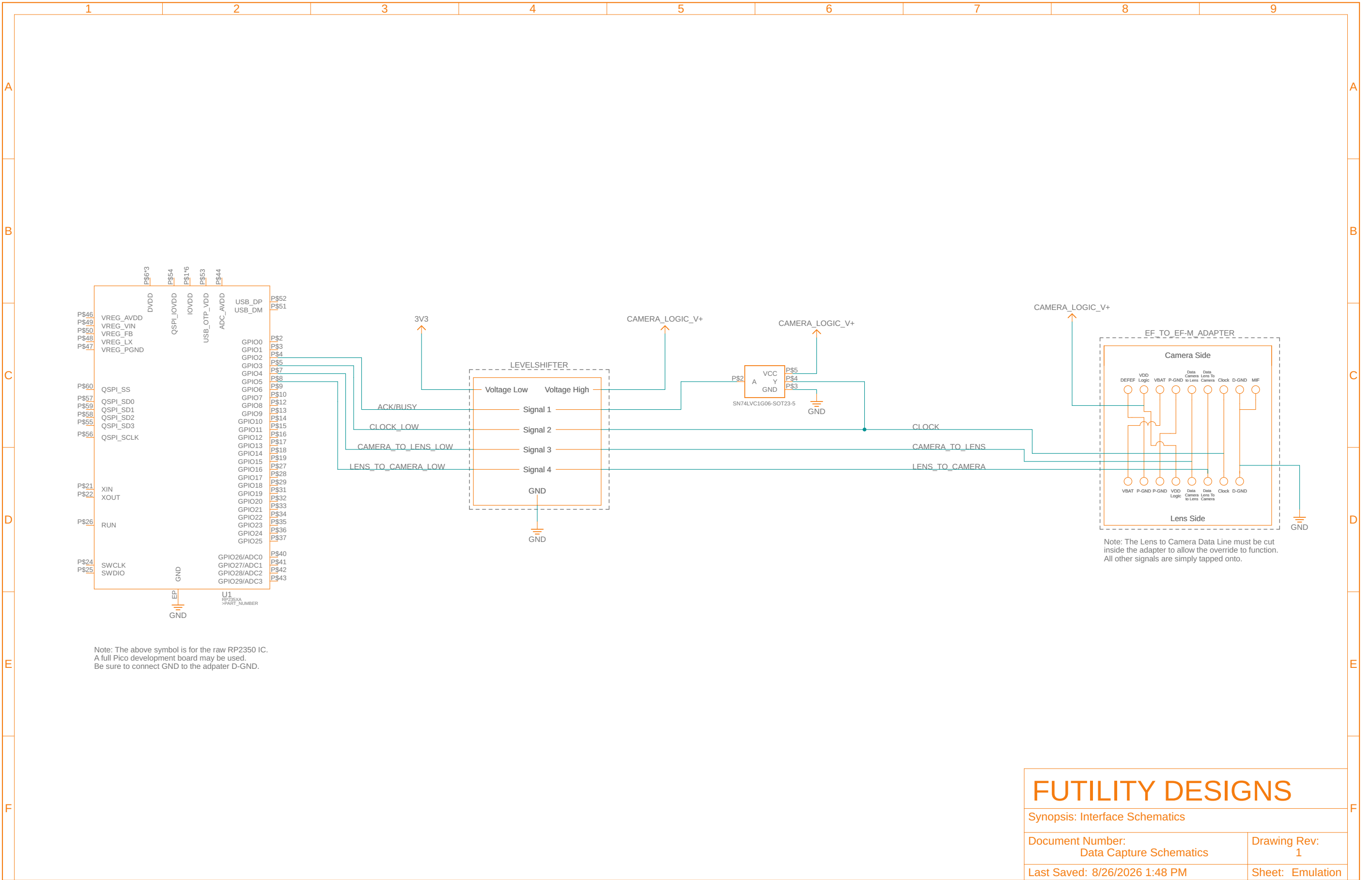
JP. (2016). *Photo-Tech*. Retrieved from Le portail à JP (Translated: The Gateway to JP): <http://jp79dsfr.free.fr/Photo-Tech/index.htm>

Appendix A – Testing Hardware Schematics

The following pages contain the schematics for each of my testing setups

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Appendix B – Testing Scripts and Programs

The following scripts and programs are planned to be published with the release of Revision 1 of this document.

EF Capture – Not Yet Uploaded

GUI based python tool used to run an assortment of data collection and testing tasks. Full documentation provided in a separate document with the program.

Excel Macros – Not Yet Uploaded

Various macros used for importing and processing data generated by the EF Capture Program.

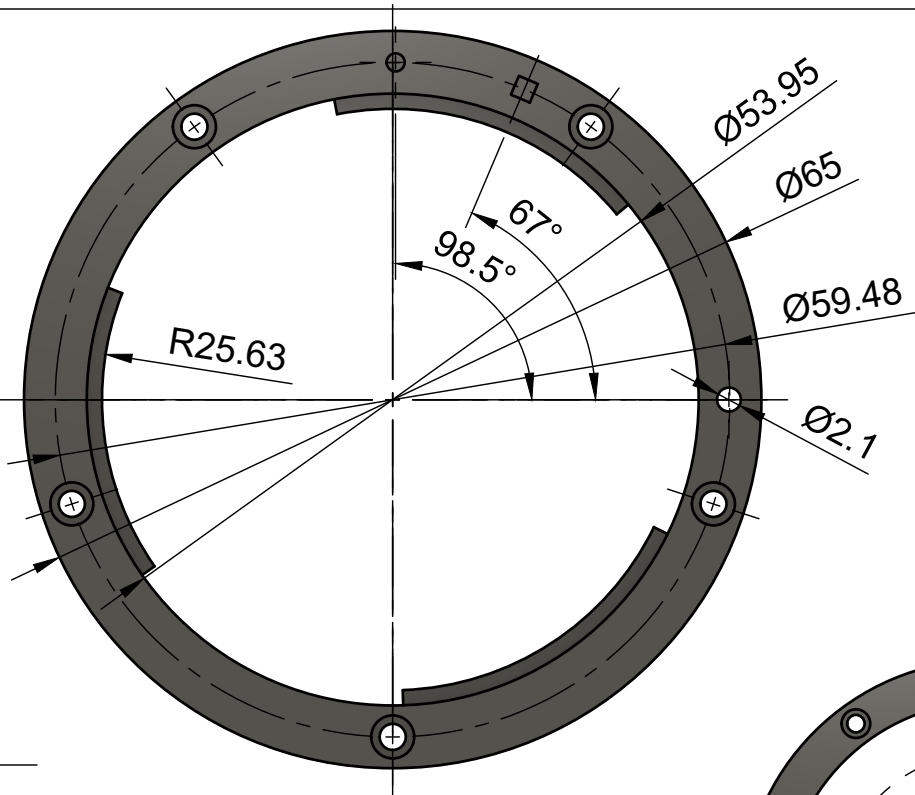
Saleae Logig2 High Level Analyzer

This high level analyzer allows for direct decoding of the Canon EF protocol in the logic2 software. It requires configuring a standard SPI analyzer first and then using that as the input to process. The analyzer is submitted to the Saleae Marketplace pending review. It is also available at:

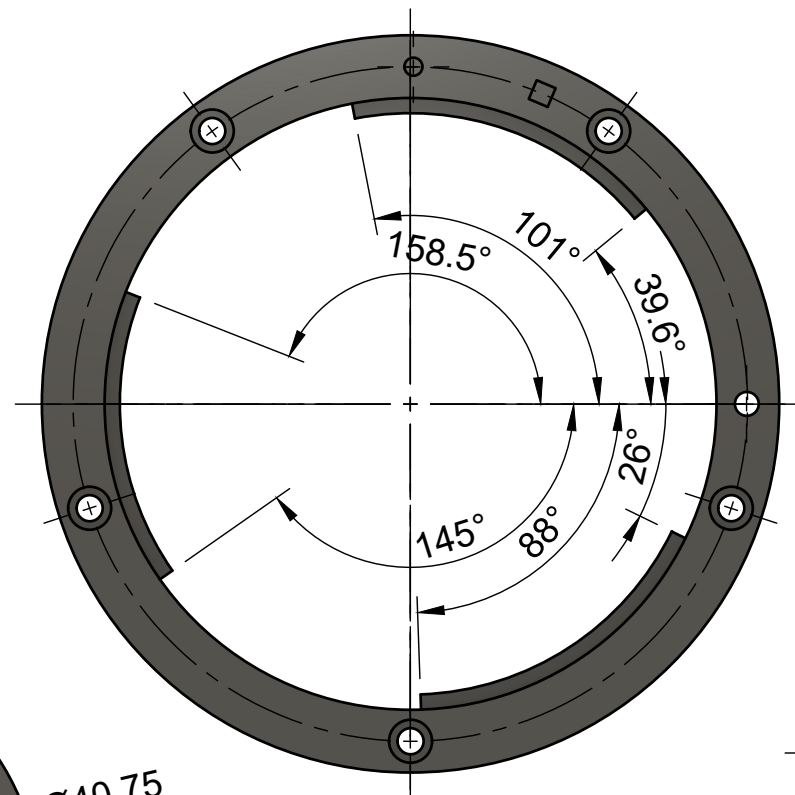
<https://github.com/FutilityDesigns/Saleae-canon-EF-High-Level-Analyzer>

Appendix C – Mechanical Drawings

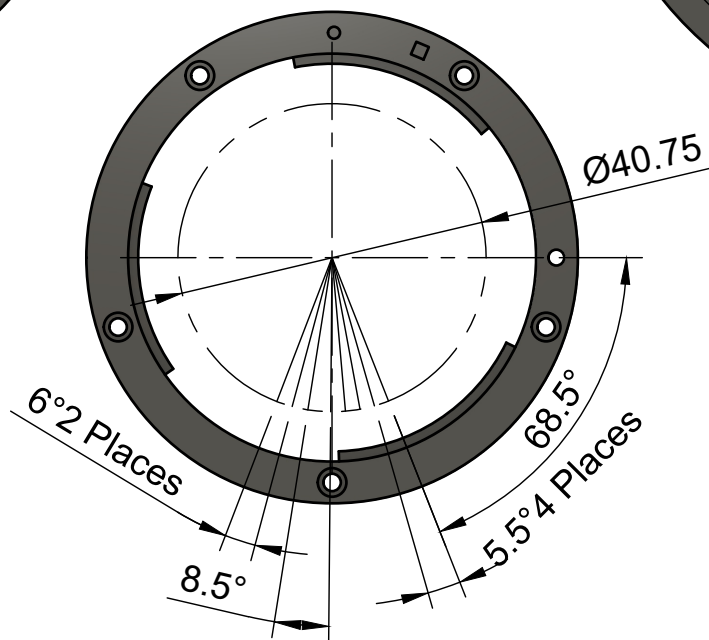
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Scale 1.5:1



Scale 1.5:1



Communication Pin Locations
Scale 1:1

Title

Canon EF Mount Flange

Created by
Chris E.

8/26/2026

Rev.
1

Date of issue
8/26/2026

Sheet
1/1